

THE EMERGENCE OF COMPLEX BEHAVIOR IN LARGE-SCALE ECOLOGICAL ENVIRONMENTS

Anonymous authors

Anonymous submission

ABSTRACT

We study how environmental scale and population size govern the emergence of increasingly complex ecological behavior in open-ended simulations. In our setting, agents receive no explicit rewards, supervision, or learning objectives; their policies change solely through reproduction, mutation, and selection, and each agent, in acting, continually reshapes both its environment and the surrounding population within a single dynamic ecology. Rather than optimizing an individual high-performance policy, we characterize how behaviors arise and propagate across large populations subject only to competition and environmental pressure. Coupling modern accelerators with a purpose-built multi-agent simulator, we scale both the environment and the population well beyond previously attempted regimes, sustaining more than 60,000 concurrent agents, each governed by its own evolved neural network policy. Within this regime we document a repertoire of emergent behaviors—long-range resource extraction, vision-mediated foraging, and predation-like interaction—that arise directly from survival and competitive pressures. By systematically varying sensing modalities and environmental scale, we establish that several of these behaviors emerge reliably only once the environment and population are sufficiently large, and that increasing scale renders the resulting behaviors markedly more stable and consistent across independent runs. Set against a long tradition of research in evolutionary and artificial-life systems, these scaling results indicate that modern hardware opens a concrete and promising avenue for studying ecology as an instrument of machine learning, in an era of increasingly abundant computation and efficient simulation frameworks. Experimental code will be released upon acceptance.

1 INTRODUCTION

Since Darwin, the evolutionary emergence of biodiversity and complex behavior has been a central object of study across a broad range of biological disciplines, spanning evolutionary theory (Darwin, 1859; Simpson, 1944), ecology (MacArthur & Wilson, 2001), and genetics (Dobzhansky, 1937). Formal accounts of these dynamics range from theories of speciation and adaptive radiation (Coyne & Orr, 2004; Schluter, 2000) to the mathematical analyses of population genetics (Fisher, 1930; Wright, 1931).

Across these fields, decades of mathematical modeling, laboratory experiments, and field work have profoundly advanced our understanding of the mechanisms of evolution in nature. Studying emergent behavior directly in the natural world nonetheless remains fundamentally constrained: as ecosystems grow larger and more complex they become correspondingly harder to control and to measure, and even where controlled experimentation is feasible, intervening at scale in wild settings risks damaging or displacing the very populations under study.

Simulation offers a complementary and controllable alternative. To examine how behavior emerges under competitive and environmental pressure, we study the open-ended evolution of neural network policies in large-scale ecological simulations. Agents are given no specified objective or reward signal; they collect resources in order to survive and to reproduce according to the dynamics of the environment, and their policies change solely through mutation as parents transmit weights to their offspring. Exploiting the rapid growth in computational resources now available to researchers, we probe these open-ended dynamics in populations of simple digital organisms at scales that were

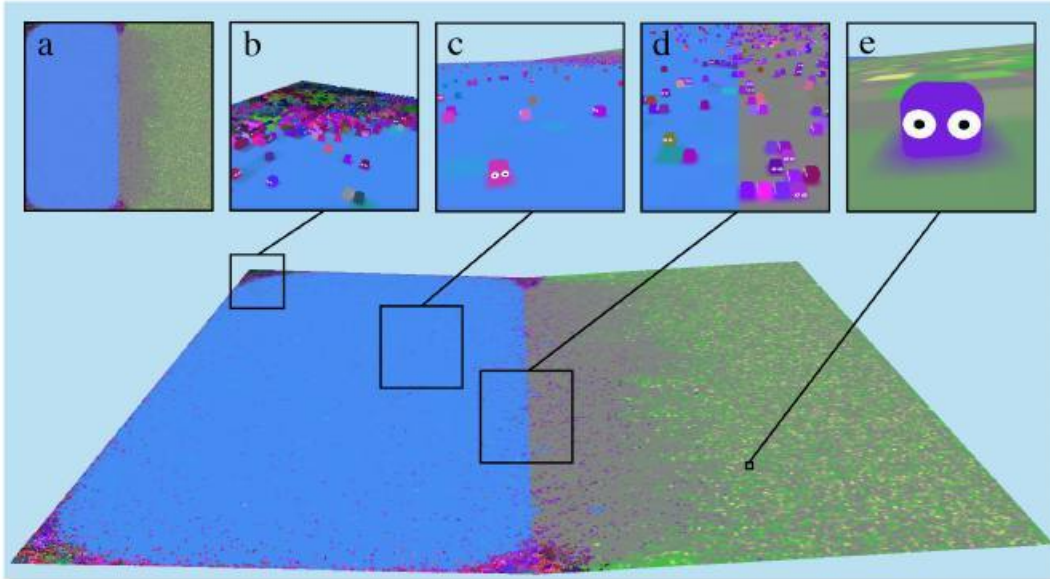


Figure 1: A 3D visualization of a 512×512 map in our environment. Inset (a) provides a top-down view of the entire map; (b) shows a group of agents clustered in a corner; (c) shows agents in the water; (d) shows agents travelling repeatedly between land and water, the movement pattern characteristic of long-range resource gathering; and (e) shows a close-up of a single agent gathering resources.

previously unattainable, and in doing so provide tools that isolate the effects of environmental scale and population size on behavioral emergence.

To this end we develop a new JAX-based simulation environment that supports rapid evaluation of large grid worlds. Our largest experiments sustain over 60,000 concurrent agents across a physical area exceeding 1,000,000 grid cells. Agents navigate the world to locate food and other resources in order to survive, and they reproduce by accumulating sufficient resources to instantiate a mutated copy of themselves. A 3D visualization of this environment is shown in Figure 1.

Our aim is twofold: to advance the understanding of ecological phenomena through controlled simulation, and to develop ecology itself into a mechanism for studying the emergence of machine behavior. Recent years have made clear that, in artificial intelligence, scale does not merely improve models but can qualitatively transform their capabilities (Wei et al., 2022a). Embodied learning, however, has largely remained confined to comparatively small environments with sparse populations. We therefore ask what behaviors emerge when scale is applied not to a single model but to an entire ecosystem. In analogy with the observation that certain reasoning behaviors appear in individual models only beyond a size threshold (Wei et al., 2022b), we characterize the strategies that evolve within populations once environmental size and complexity become sufficiently rich.

Through extensive experiments, we identify several regimes in which sensor configuration and environmental scale jointly and decisively shape the behaviors that emerge, with lasting consequences for population dynamics. Agents equipped with a simple compass sensor reliably develop long-range resource-gathering expeditions, a behavior that emerges far less consistently at small scales; agents additionally endowed with vision develop markedly more effective foraging, together with more precise attacking behavior, and this advantage too is realized more reliably as scale increases. Taken together, these results establish that the scale and complexity afforded by modern hardware accelerators constitute a powerful and practical approach to studying how evolution in ecological environments gives rise to new behaviors.

2 RELATED WORK

The emergence of life has inspired computational research since the field’s inception. Conway’s Game of Life (Games, 1970) demonstrated that populations governed by simple local rules can give rise to strikingly complex global dynamics, while early systems such as Tierra (Ray, 1992) and Avida (Brown) modeled life as self-replicating programs that evolve through mutation. In a kindred spirit, Reynolds (1987) established that local rules of motion alone suffice to produce emergent flocking and schooling.

Building on these principles, researchers have evolved both the locomotive morphology (Sims, 1994a,b; Ha, 2018; Gupta et al., 2021; Auerbach et al., 2014; Auerbach & Bongard, 2012; Jeon et al., 2025; Auerbach & Bongard, 2014; Szerlip & Stanley, 2013) and the sensing apparatus (Pratt et al., 2022; Mugan & MacIver, 2020; Tiwary et al., 2025) of virtual agents, while neuroevolution methods evolve neural network architectures directly (Stanley & Miikkulainen, 2002, 2004; Stanley et al., 2009). Our objective differs in kind: rather than seeking agents that adapt to their environment under a reward or objective signal, we characterize what behaviors emerge purely as a consequence of ecological pressure, and how environmental scale and complexity govern those behaviors.

Evolution has likewise informed the internal dynamics of optimization algorithms. Genetic algorithms (Holland, 1992; Jong, 1975) and evolutionary strategies (Wierstra et al., 2014; Hansen & Ostermeier, 2001; Rechenberg, 1973; Schwefel, 1995; Conti et al., 2018) recast the principles of natural selection as optimization procedures, while related methods explicitly reward novelty among the resulting solutions (Lehman & Stanley, 2011; Cully et al., 2015). Such approaches have been used to optimize neural networks for tasks such as Atari games and Mujoco locomotion tasks (Salimans et al., 2017; Such et al., 2017; Mania et al., 2018). Although we too employ evolutionary mechanisms to update agents, our objective is again distinct: we study emergent behavior itself, rather than developing a new optimization algorithm or applying an existing one to a fixed objective.

A rich line of work draws on biological inspiration to develop organisms situated within an environment. Early examples include Bedau (1992), Yaeger et al. (1994), Channon & Damber (1998), and Tisue et al. (2004). More recent efforts use cellular automata to grow organisms with emergent morphologies and behaviors (Mordvintsev et al., 2020; Robinson, 2021; Heinemann, 2021; Randazzo & Mordvintsev, 2023; Caussan, 2025), while Mordatch & Abbeel (2018) and Park et al. (2023) employ simulated environments to investigate the emergence of language and communication among agents. The emergence of behavior through autocurricula (Leibo et al., 2019)—the principle that a population of agents constitutes a progressively more challenging learning landscape for its own members—has likewise been examined, though in environments with substantially smaller populations than ours (Leibo et al., 2018; Team et al., 2021). Complementarily, Wang et al. (2019) and Aki et al. (2024) pursue the reverse direction, evolving environments explicitly optimized to elicit effective agent learning. Although these works evolve the environment rather than the agent, they directly motivate ours by demonstrating how environmental complexity can elicit increasingly sophisticated behavior.

Most closely related to our work are efforts that simulate populations of agents interacting within complex environments. Baker et al. (2019) study a Hide-and-Seek task in which a small number of hider agents learn to exploit environmental tools to evade seekers, both trained via reinforcement learning. Other works similarly train populations for specific tasks—playing video games (Jaderberg et al., 2019), manipulating objects (Petrenko et al., 2021), or capturing agents (Zheng et al., 2017; Yamada et al., 2020). Although many of these efforts probe emergence, they do so under fixed tasks and predefined objectives. The Neural MMO framework (Suarez et al., 2019, 2023) adopts a more open-ended form of reinforcement learning, situating agents in a massively multi-agent game and optimizing for survival rather than a fixed objective; its population, however, is capped at a fixed count (experiments consider up to 128 concurrent agents) and lacks generational inheritance, which limits the study of population dynamics. Hamon et al. (2023) aligns more closely with our setting through variable population sizes, genetically inspired evolution, and generational timelines, but its environment is considerably smaller and less feature-rich, and the role of scale is not itself an object of study. Lu et al. (2024) likewise simulates agents that evolve to gather resources, yet also operates at a far smaller scale (at most 256 agents) and emphasizes tool use through programmable robots. Each of these works takes a meaningful step toward examining emergent ecological dynamics; a sys-

tematic account of scale, however, requires populations that grow and shrink naturally, evolve without predefined objectives, and interact within environments rich enough to support diverse strategies at sizes where collective patterns can emerge. Our environment is the first to combine all of these elements at once, and it is precisely this combination that renders accessible the scale-dependent ecological phenomena we study here.

Finally, our approach is informed by recent work emphasizing environmental speed and scale, which employs optimized libraries to raise computational throughput and the number of agents that can interact within game-like environments (Koyamada et al. 2023, Matthews et al. 2024, Lechner et al. 2023). We build directly on this progress in computational efficiency, adopting comparable JAX-based acceleration techniques to attain large scale and fast simulation in the ecological setting.

3 METHOD

3.1 ECOLOGICAL GAMES

We formalize the interaction between populations of agents and their environment as a class of multiplayer game that we term an **Ecological Game (EG)**. Like the Markov decision processes (MDPs) of reinforcement learning and the stochastic games (SGs) of multi-agent reinforcement learning (see Sutton et al. (1998) and Albrecht et al. (2024) for reference), an EG possesses underlying Markovian transition dynamics; unlike them, it is expressly designed to represent changing populations of agents in an objective-free setting. In common with an SG, an EG comprises a set of agents I , a state space S , an initial state distribution ρ_0 , an action space A , and a transition function T that maps a state s_t together with a joint action $a_{t,i}$ for each player to a distribution over successor states s_{t+1} . It departs from the SG in two respects: it omits the reward function entirely, and it introduces a population function $\Gamma(s_t)$ that identifies which agents in I are currently alive together with the identity of their parents, so that birth and death events can be inferred. Just as MDPs and SGs admit partially observable counterparts—POMDPs and POSGs—an EG admits a partially observable variant, which we term a POEG. In a POEG no agent observes the full state; instead, an observation function $O(s_t, i)$ maps a state and a player to that player’s observation. Our environment is precisely such a POEG, with the observation function fixed by the sensor configuration detailed in the following subsection.

3.2 ENVIRONMENT

The specific POEG used throughout our experiments is a large grid world populated with resources that agents must collect in order to survive and reproduce. The environment maintains a factored state space $s_t = \{s_t^m, s_t^a\} \in S = S^m \times S^a$, in which s_t^m denotes the 2D map-level data—terrain together with the spatial distribution of resources—and s_t^a denotes the per-agent data, comprising position, health, and internal resources.

Terrain: Each map is specified by a rock layer that assigns a height to every grid cell. Water is initialized into all cells whose rock value lies below a preset sea level and thereafter flows downhill according to a discrete flow simulation driven by this height field. Our experiments employ the six global terrain configurations shown in Figure 2—**Ocean, Beach, Island, Lake, Isthmus**, and **Channel**—instantiated at grid sizes of 64×64 , 128×128 , 256×256 , 512×512 , and 1024×1024 . Whenever the grid size is changed, the initial population is scaled in proportion to the change in total grid area; once simulation begins, the population thereafter fluctuates freely according to the dynamics of the environment.

Resources and Agent Health: To survive and reproduce, agents must collect three resources—**water, energy**, and **biomass**. Biomass and energy are initially distributed at random across the environment, and both may subsequently be consumed and redistributed by individual agents.

Every action expends a small quantity of energy and water. Spent water is returned to the landscape and flows downhill toward the nearest body of water, whereas spent energy is consumed and destroyed. Biomass serves as the currency of reproduction: each agent requires a fixed amount of biomass merely to exist, and cannot reproduce until it has accumulated enough surplus to endow its offspring. As with water, total biomass is conserved over the simulation; unlike water, however,

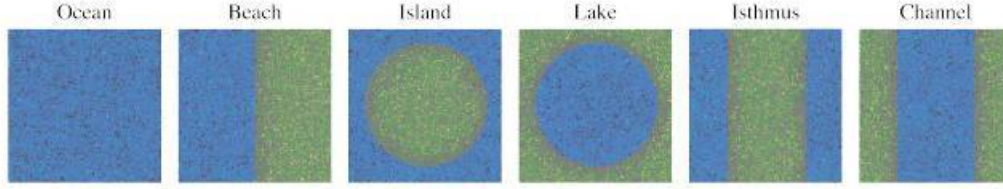


Figure 2: Overhead views of the six terrains used in our experiments, shown at the 256×256 grid size. Small dark spots denote individual agents, while green and yellow indicate concentrations of energy and biomass, respectively. Each image was captured 5000 steps into an episode—after most biomass in the water has been consumed, but before agents have evolved to survive on land for extended periods.

biomass neither flows downhill nor moves of its own accord. Energy, in contrast, is not conserved, accruing slowly on free-standing biomass through a photosynthesis process. These resource dynamics are deliberately chosen to pose a more demanding problem than the collection of a single food source, requiring agents to balance three interdependent quantities. Ablations on the resource rules are reported in Appendix F.2

Agents additionally carry health points (HP), which they must maintain to avoid death. Overspending a critical resource causes an agent’s health to deteriorate rapidly, although it may expend energy and water to recover HP slowly over time. Agents may also attack one another, reducing the target’s HP, and each incurs a small but monotonically increasing amount of damage as a function of age, so that—absent any other cause of death—every agent eventually dies of old age. When an agent’s HP reaches zero it dies, depositing whatever resources it was carrying into the grid cell it occupied; an agent that attacks and kills another may then consume the victim’s deposited resources, a mechanism that affords a rudimentary basis for predation. Ablations on the HP hyperparameters are reported in Appendix F.8

Sensors: Agents perceive the environment through four distinct sensors. The **internal** sensor reports the agent’s age, health, and carried resources; the **external** sensor reports the resources present in the agent’s own grid cell; the **compass**, inspired by the magnetoreception observed in birds and other animals, encodes the agent’s global heading as a 4-way one-hot vector; and **vision** supplies a 7×7 top-down rendered patch centered on the agent, comprising three color channels together with a fourth channel that encodes the local elevation gradient. Ablations on the vision configuration are reported in Appendix F.6

Actions: Agents select from a finite set of discrete actions organized into five categories. The **Rest** action performs no operation and incurs the lowest energy cost. Three **Move** actions are available—rotate clockwise, rotate counterclockwise, and advance one grid cell forward—subject to an exclusion constraint: because no two agents may occupy the same cell, a move is canceled whenever an agent attempts to enter an occupied cell, or whenever two agents attempt to enter the same cell simultaneously. A single **Attack** action kills every agent within the 3×3 square directly in front of the actor; ablations on attack strength are reported in Appendix F.7 Three distinct **Eat** actions, one per resource type, each transfer a fixed amount of the corresponding resource from the environment into the agent’s stomach—which has a fixed maximum capacity—whenever the agent shares a cell with that resource.

Finally, a single **Reproduce** action produces an offspring through single-parent reproduction. This action succeeds only once the agent has accumulated sufficient resources to endow its child. The offspring is instantiated in the cell directly behind the parent, its policy is a mutated copy of the parent’s weights, and it immediately inherits a fixed portion of the parent’s resources.

Throughout our experiments we selectively disable specific sensors, or the attack action, in order to isolate the ecological consequences of endowing agents with different capabilities. We consider three sensing combinations—internal+external resources only (**R**), internal+external+compass (**RC**), and internal+external+compass+vision (**RCV**)—and append **+A** to indicate that attacking is enabled; thus, for example, (**RC+A**) denotes an agent equipped with resource sensors and a compass that is additionally permitted to attack.

3.3 POLICIES

Each agent policy is realized as a small, memoryless multilayer perceptron (MLP). We adopt memoryless agents deliberately, so as to simplify the computational model and to reduce the per-agent parameter count. The raw values from each sensor are flattened, normalized to the range $[-1, 1]$, and passed through separate linear layers of output dimension 64; the resulting embeddings are summed and fed through a 2-layer MLP with hidden dimension 64 and ReLU nonlinearities. Because the action space is discrete, the final layer is linear with output dimension equal to the number of available actions, and actions are sampled from a softmax distribution whose temperature is itself allowed to evolve alongside the network weights. Weights are stored in half-precision bfloat16 to support larger populations and faster runtimes. Depending on the active sensor set, each agent’s network contains roughly 10k–25k parameters. Ablations on network size and architecture are reported in Appendix F.5

Across all runs, the weights of the initial population are drawn by Kaiming initialization (He et al., 2015) and bias vectors are set to zero. The policy of an agent born mid-episode is formed by copying its parent’s weights and perturbing every mutable weight with additive Gaussian noise of standard deviation 3×10^{-2} , a mutation scheme consistent with prior neuroevolution research (Such et al., 2017). Although more elaborate operators such as crossover are available, we adopt this minimal perturbation deliberately, keeping reproduction as simple as possible so as to isolate the effect of environmental scale within a controlled setting; the consequences of alternative mutation magnitudes are examined in Appendix F.4. Empirically, the low-precision floating-point representation required a somewhat larger mutation rate than would otherwise be expected.

Each agent additionally carries a 3-channel trait that determines its rendered color on the map. This color trait mutates in the same manner as the network weights and fixes the agent’s appearance to any other agent equipped with the vision sensor, in principle enabling agents to discriminate among distinct classes of neighbors.

3.4 COMPUTATION AND REPRODUCIBILITY

Our experiments are implemented end-to-end in JAX (Bradbury et al., 2018) and executed on Nvidia H100 and A40 GPUs within a purpose-built, GPU-native simulation environment. Unless otherwise noted, each experiment runs for 2M environment steps or until extinction, with per-seed wall-clock times ranging from under an hour on 64×64 grids to approximately 10 hours on 1024×1024 grids using a single H100 GPU. Appendix E details the simulator architecture, reports additional scaling measurements, and discusses the reproducibility considerations that follow from running large, heavily randomized simulations on modern accelerators.

4 EXPERIMENTS

The experiments that follow establish how sensing modality, environmental scale, and population size jointly shape the behaviors that emerge in our simulations. We first characterize long-distance resource gathering (Section 4.1), then turn to vision-mediated foraging and predation-like interaction (Section 4.2), and finally summarize the ablations that establish the robustness of these findings (Section 4.3).

4.1 LONG-DISTANCE RESOURCE GATHERING

Our first experiment examines the capacity of agents to travel long distances inland in search of resources. We find that in environments with clear directional structure, agents equipped with a compass (**RC**) consistently develop substantially more effective long-range resource-gathering behavior than the baseline class (**R**) that lacks a compass, and that this behavior emerges both more reliably and more effectively as environmental scale increases.

Consider first a representative trajectory of the (**R**) agent class, shown in Figure 3. By 5000 time steps (a), every initial agent on dry land has perished for want of water, while those in the water have proliferated rapidly on the abundant initial resources; at this early stage the population has not yet evolved policies capable of efficiently locating resources. From this initial peak, agents in the

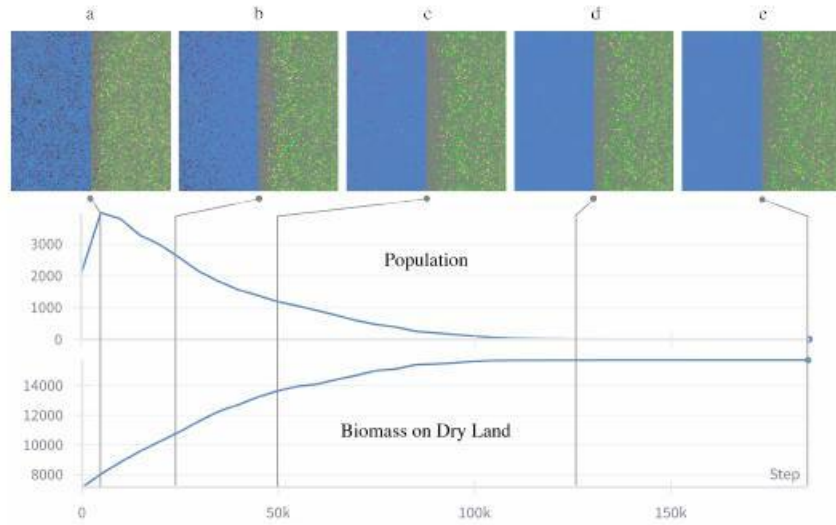


Figure 3: A representative trial with **(R)** agents in a 256×256 **Beach** environment. The top chart plots population over time; the bottom chart plots the free biomass on dry land. The labeled points (a)–(e) mark the stages discussed in the text.

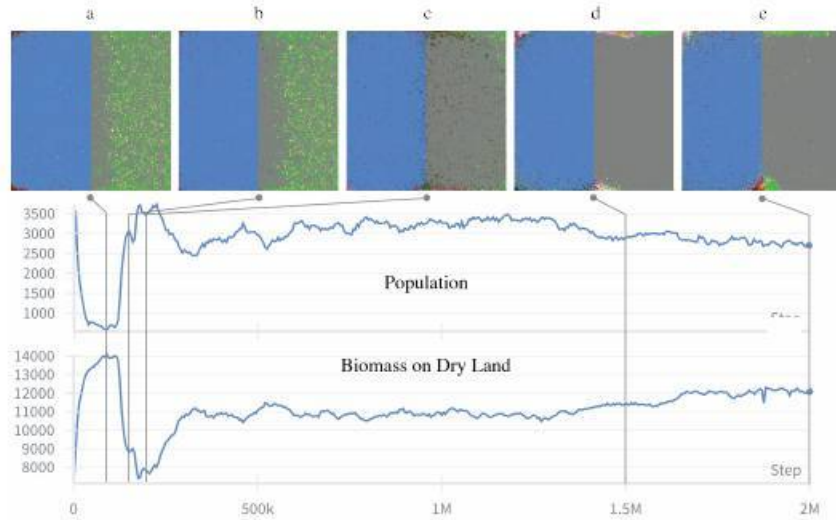


Figure 4: A representative trial with **(RC)** agents in a 256×256 **Beach** environment. The top chart plots population over time; the bottom chart plots the free biomass on dry land. In contrast to Figure 3, the compass enables a sustained mining strategy and the population persists for the full run.

water occasionally wander onto land (b), where they typically fail to find their way back to the ocean and die, depositing their biomass ashore. By 50,000 time steps (c), this gradual process has nearly doubled the uneaten biomass on land. The resulting transfer of resources from water to land drives the population down, as the water no longer holds enough biomass to sustain a large population; by 125,000 steps (d) almost all biomass has migrated onto land and the population is on the verge of collapse, with the final agent dying at approximately 185,000 steps (e).

This trajectory illustrates a central ecological consequence of the agents' inability to return to water: individual deaths not only remove agents from the population but also reshape the resource distribution in ways that determine the fate of the entire population. The same catastrophe, however, creates an opportunity—an opening for any agent that discovers how to walk onto land, forage the rich biomass deposited by its dead ancestors, and safely return to water.

Contrast this with the trajectory in Figure 4 in which agents are additionally equipped with a compass (RC). The early dynamics are similar (a): the population contracts as free biomass accumulates rapidly on land. Here, however, some agents exploit the compass to drift into the relative safety of the corners. Around 150,000 steps (b), agents begin to “mine” the biomass on land, using the compass to navigate repeatedly between land and water; by 200,000 steps (c) these miners have cleared the entire beach. Thereafter the population fluctuates only mildly, with agents forming clumps around the edges (d), and remains largely stable for the full 2,000,000 time steps (e). The compass thus converts an otherwise terminal ecological catastrophe into a sustained, self-perpetuating foraging strategy.

Table 1: Mining events in the **Beach** world for populations of agents with a compass (RC) versus without a compass (R), across five world sizes and initial populations.  denotes the number of seeds exhibiting a mining event, and  the number of seeds in which the population went extinct before 2M steps. Equipping agents with a compass yields consistent mining, whereas its absence yields almost none; mining also becomes more reliable, and extinction rarer, as world size grows, occurring in every seed at and above the 256×256 scale for (RC) agents. See Table 3 in Appendix A for per-seed detail.

	1024 × 1024		512 × 512		256 × 256		128 × 128		64 × 64	
	32768		8192		2048		512		128	
										
(RC)	4/4	0/4	4/4	0/4	4/4	0/4	3/4	1/4	2/4	3/4
(R)	1/4	0/4	0/4	4/4	0/4	4/4	0/4	4/4	0/4	4/4

We quantify these observations across a full suite of **Beach** experiments spanning grid sizes 1024 × 1024, 512 × 512, 256 × 256, 128 × 128, and 64 × 64, with initial populations of 32768, 8192, 2048, 512, and 128 agents, respectively, each configuration run with four random seeds. For each run we record whether mining behavior emerges—operationalized as a drop of at least 10% in the free biomass on dry land relative to its historical maximum—together with the onset time of the mining event, defined by the local maximum in land biomass, and the associated percentage drop. Mining and extinction events are summarized in Table 1 with per-seed detail in Table 3 of Appendix A. The pattern is clear and consistent: mining emerges in nearly every (RC) run but only rarely in (R) runs, and it becomes more reliable as the environment grows, occurring in every seed at and above the 256×256 scale. At smaller scales, by contrast, populations are markedly less stable and substantially more prone to early extinction.

To characterize these effects more precisely, and to place them on firmer statistical footing, we run additional seeds with (R) and (RC) agents at the 512×512 world size (32 seeds) and the 128×128 world size (64 seeds), and apply a Kaplan–Meier estimator to the probabilities of extinction and of mining as functions of simulator time. Extinction is rarer and slower to occur at larger world sizes, while mining is more frequent and occurs sooner at smaller grid sizes. A complementary Cox proportional hazards model, fit jointly across world sizes and compass usage, likewise establishes that larger world size substantially reduces the extinction hazard while enabling far more rapid mining discovery than smaller worlds afford, and that enabling the compass produces an effect of comparable magnitude. Full details are provided in Appendix B.

To assess how terrain configuration modulates the emergence of mining, we run analogous experiments with (R) and (RC) agents in the **Island**, **Lake**, **Isthmus**, and **Channel** terrains at the 512×512 grid size. Mining and extinction events are reported in Table 2 with per-run detail in Table 4 of Appendix A. The pattern established above holds across every terrain: equipping agents with a compass yields earlier and more consistent mining, together with more stable populations and lower extinction rates. (RC) agents adapt more readily to the island and isthmus than to the lake and channel, where they occasionally go extinct—a difference we attribute to a simple navigational invariant present in the former terrains, where walking due east or west always returns an agent to water, and absent in the latter. The (R) agent, by contrast, cannot reliably adapt to any of these terrains, though it struggles least on the island. Figure 12 in Appendix D visualizes mining in the island terrain.

Table 2: Mining and extinction events in the **Island**, **Lake**, **Isthmus**, and **Channel** terrains for populations of agents with a compass (**RC**) versus without a compass (**R**), at the 512×512 grid size.  denotes the number of seeds exhibiting a mining event, and  the number of seeds in which the population went extinct before 2M steps. Across every terrain the compass yields earlier and more consistent mining together with lower extinction rates, confirming that the effect is not specific to the **Beach**. See Table 4 in Appendix A for per-seed detail.

	Island		Lake		Isthmus		Channel	
								
(RC)	4/4	0/4	2/4	2/4	4/4	0/4	3/4	1/4
(R)	3/4	1/4	2/4	4/4	1/4	3/4	0/4	4/4

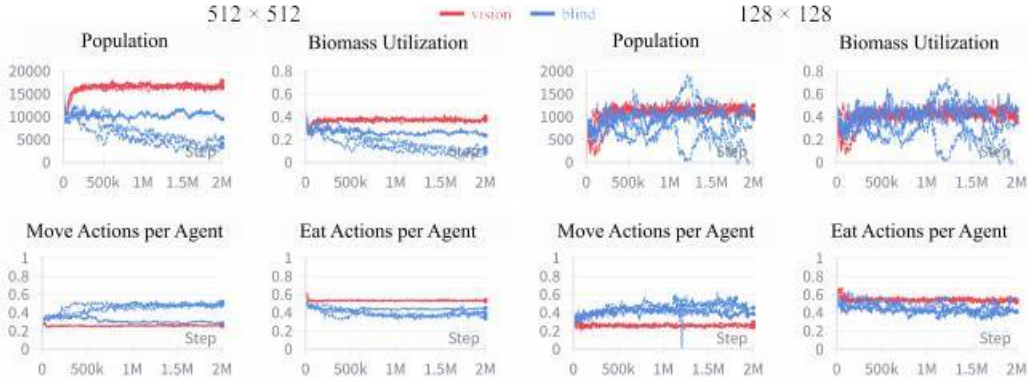


Figure 5: Population, biomass utilization, move actions per agent, and eat actions per agent in 512×512 and 128×128 **Ocean** worlds. Biomass utilization, move actions per agent, and eat actions per agent are averaged across agents at each time step. Each panel shows four seeds with vision agents (**RCV + A**) (red) and four seeds with blind agents (**RC + A**) (blue). With vision, populations are larger and more stable and biomass utilization is higher, driven by reduced movement together with more frequent and more effective eating. At larger scales these dynamics are more stable, exhibiting less variance across runs, which indicates that vision confers a growing advantage over blindness for adapting foraging behavior as the environment expands. See Figure 9 in Appendix C for the comparison of grid sizes 1024×1024 and 256×256 .

4.2 VISUAL FORAGING AND PREDATION

We next turn to the **Ocean** environment to determine whether agents can adapt to exploit vision, both for locating free resources and for attacking other agents. We compare (**RCV+A**) agents, equipped with resource sensing, a compass, and a 7×7 visual sensor, against (**RC+A**) agents that possess only resource and compass sensors, holding the attack capability fixed across the two classes so that any difference is attributable to vision alone.

Figure 5 plots population, biomass utilization, movement actions, and eat actions at the 512×512 and 128×128 grid sizes; the corresponding plots for the remaining grid sizes appear in Figure 9 of Appendix C and summary statistics in Table 5 of Appendix A. Biomass utilization is defined as the fraction of the environment’s total biomass currently held within agents. In runs with vision (red curves), agents spend less time moving and more time eating, sustain larger populations, and attain greater biomass utilization—a coherent signature that agents have adapted to leverage vision for more efficient foraging. Across environmental scales, the variance between runs decreases substantially in larger environments, and runs equipped with vision exhibit markedly more stable dynamics, with less variation in these metrics over time; the foraging advantage conferred by vision therefore grows more pronounced, and more reliable, as the environment is enlarged.

Figure 6 reports the averaged attacks per agent and homicides per attack for these runs. Vision agents attack relatively infrequently yet sustain a high kill rate of approximately 75%, a combination consistent with precise, targeted attacking rather than indiscriminate aggression. Blind agents in large grids occasionally attain high kill rates as well, but with far less stable dynamics, and their elevated rates are more difficult to interpret: visual inspection (Figure 13 in Appendix D) shows that they often coincide with dense clusters of agents forming at a single location—clusters that almost never arise in runs with vision. Under such crowding, incidental contact alone can inflate the kill rate, so we regard the vision-conditioned signal, rather than the blind agents’ sporadic spikes, as the more reliable evidence of targeted attacking; the persistent clustering of blind agents further suggests that vision additionally helps agents adapt to avoid one another. Taken together, these observations point to a vision-mediated form of predation-like interaction, though our current measurements do not by themselves separate deliberate pursuit from opportunistic killing. In smaller environments the contrast between vision and blind agents is less pronounced, yet populations of vision agents remain considerably more stable.

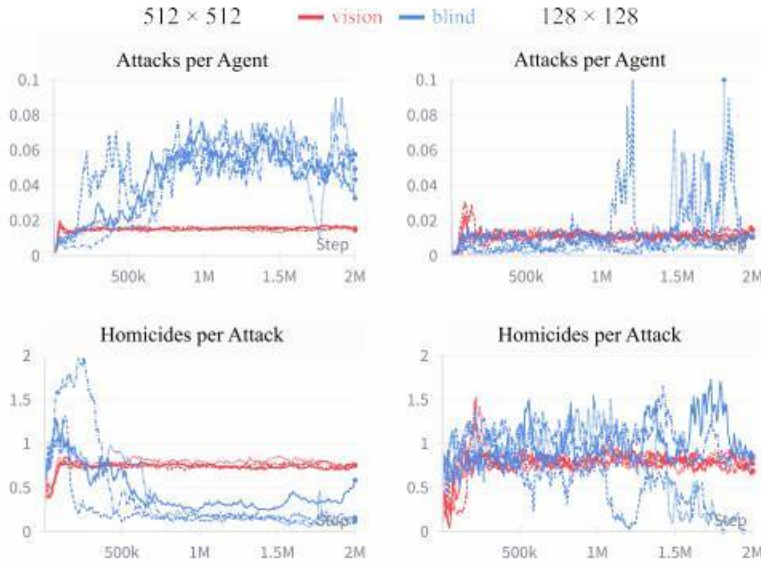


Figure 6: Attacks per agent and homicides per attack, averaged across agents, in 512×512 and 128×128 **Ocean** worlds. Each panel shows four seeds with vision agents (**RCV + A**) (red) and four seeds with blind agents (**RC + A**) (blue). With vision, agents attack less frequently but with substantially higher precision, consistent with a targeted, predation-like use of the attack action. This dynamic is more pronounced at larger scales. See Figure 10 in Appendix C for the comparison of grid sizes 1024×1024 and 256×256 .

4.3 ABLATIONS

To establish the robustness of these behaviors to simulator configuration, Appendix F reports a comprehensive set of ablations over initial population size (F.1), resource rules (F.2), mutation rate (F.4), agent network configuration (F.5), vision range (F.6), attack strength (F.7), and agent HP (F.8). Collectively, these sweeps confirm that the phenomena reported above are not artifacts of any single hyperparameter choice. Appendix F.3 additionally presents a dedicated experiment establishing that the dynamics observed in larger environments cannot be attributed merely to self-averaging over many smaller ones.

5 CONCLUSION

Our study establishes open-ended ecological simulation as a powerful and practical paradigm for investigating how complex behavior emerges under evolutionary pressure. Across experiments in worlds of varying scale and complexity, we show that environmental scale and sensory richness

are decisive determinants of whether long-range foraging, vision-mediated feeding, and targeted attacking reliably develop from survival and competition alone. Taken together, these results indicate that ecological and evolutionary pressures, when coupled with sufficiently large and rich environments, constitute a viable pathway toward progressively more capable artificial agents. Critically, our experiments further demonstrate that such behaviors can be evolved in large populations within a modest hardware budget and entirely without explicit objectives. We anticipate that these findings will motivate further exploration of ecology as a mechanism for both developing and understanding emergent behavior in artificial agents.

REFERENCES

- Fuma Aki, Riku Ikeda, Takumi Saito, Ciaran Regan, and Mizuki Oka. Llm-poet: Evolving complex environments using large language models. In *Proceedings of the Genetic and Evolutionary Computation Conference Companion*, pp. 243–246, 2024.
- Stefano V. Albrecht, Filippos Christianos, and Lukas Schäfer. *Multi-Agent Reinforcement Learning: Foundations and Modern Approaches*. MIT Press, 2024. URL <https://www.marl-book.com>
- Joshua Auerbach, Deniz Aydin, Andrea Maesani, Przemyslaw Kornatowski, Titus Cieslewski, Grégoire Heitz, Pradeep Fernando, Ilya Loshchilov, Ludovic Daler, and Dario Floreano. Robogen: Robot generation through artificial evolution. In *Artificial Life Conference Proceedings*, pp. 136–137. MIT Press One Rogers Street, Cambridge, MA 02142-1209, USA journals-info ..., 2014.
- Joshua E Auerbach and Josh C Bongard. Environmental influence on the evolution of morphological complexity in machines. *PLoS computational biology*, 10(1):e1003399, 2014.
- Joshua E Auerbach and Joshua C Bongard. On the relationship between environmental and morphological complexity in evolved robots. In *Proceedings of the 14th annual conference on Genetic and evolutionary computation*, pp. 521–528, 2012.
- Bowen Baker, Ingmar Kanitscheider, Todor Markov, Yi Wu, Glenn Powell, Bob McGrew, and Igor Mordatch. Emergent tool use from multi-agent autocurricula. In *International conference on learning representations*, 2019.
- Chris Bamford, Shengyi Huang, and Simon Lucas. Griddly: A platform for ai research in games. *arXiv preprint arXiv:2011.06363*, 2020.
- Mark Bedau. Measurement of evolutionary activity, teleology, and life. 1992.
- James Bradbury, Roy Frostig, Peter Hawkins, Matthew James Johnson, Chris Leary, Dougal Maclaurin, George Necula, Adam Paszke, Jake VanderPlas, Skye Wanderman-Milne, and Qiao Zhang. JAX: composable transformations of Python+NumPy programs, 2018. URL <http://github.com/jax-ml/jax>
- Chris Adami and C Titus Brown. Evolutionary learning in the 2d artificial life system avida.
- Léo Caussan. The bibites: Digital life, 2025. URL <https://www.thebibites.com/> Artificial life simulation software / game.
- AD Channon and RI Damper. Perpetuating evolutionary emergence. 1998.
- Edoardo Conti, Vashisht Madhavan, Felipe Petroski Such, Joel Lehman, Kenneth Stanley, and Jeff Clune. Improving exploration in evolution strategies for deep reinforcement learning via a population of novelty-seeking agents. *Advances in neural information processing systems*, 31, 2018.
- Jerry A. Coyne and H. Allen Orr. *Speciation*. Sinauer Associates, Sunderland, MA, 2004.
- Antoine Cully, Jeff Clune, Danesh Tarapore, and Jean-Baptiste Mouret. Robots that can adapt like animals. *Nature*, 521(7553):503–507, 2015.

- Charles Darwin. *On the Origin of Species*. John Murray, London, 1859.
- Theodosius Dobzhansky. *Genetics and the Origin of Species*. Columbia University Press, New York, 1937.
- Ronald Aylmer Fisher. *The Genetical Theory of Natural Selection*. Clarendon Press, Oxford, 1930.
- Mathematical Games. The fantastic combinations of john conway’s new solitaire game “life” by martin gardner. *Scientific American*, 223:120–123, 1970.
- Agrim Gupta, Silvio Savarese, Surya Ganguli, and Li Fei-Fei. Embodied intelligence via learning and evolution. *Nature communications*, 12(1):5721, 2021.
- David R Ha. Reinforcement learning for improving agent design. *Artificial Life*, 25:352–365, 2018. URL <https://api.semanticscholar.org/CorpusID:52945447>
- Gautier Hamon, Eleni Nisioti, and Clément Moulin-Frier. Eco-evolutionary dynamics of non-episodic neuroevolution in large multi-agent environments. In *Proceedings of the Companion Conference on Genetic and Evolutionary Computation*, pp. 143–146, 2023.
- Nikolaus Hansen and Andreas Ostermeier. Completely derandomized self-adaptation in evolution strategies. *Evolutionary computation*, 9(2):159–195, 2001.
- Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Delving deep into rectifiers: Surpassing human-level performance on imagenet classification. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pp. 1026–1034, 2015.
- Christian Heinemann. Alien. <https://github.com/chrxh/alien> 2021.
- John H. Holland. Adaptation in natural and artificial systems: An introductory analysis with applications to biology, control, and artificial intelligence. 1992. URL <https://api.semanticscholar.org/CorpusID:58781161>
- Max Jaderberg, Wojciech M Czarnecki, Iain Dunning, Luke Marris, Guy Lever, Antonio Garcia Castaneda, Charles Beattie, Neil C Rabinowitz, Ari S Morcos, Avraham Ruderman, et al. Human-level performance in 3d multiplayer games with population-based reinforcement learning. *Science*, 364(6443):859–865, 2019.
- Hyeonseong Jeon, Ainaz Eftekhari, Aaron Walsman, Kuo-Hao Zeng, Ali Farhadi, and Ranjay Krishna. Convergent functions, divergent forms. *arXiv preprint arXiv:2505.21665*, 2025.
- Kenneth A. De Jong. An analysis of the behavior of a class of genetic adaptive systems. 1975. URL <https://api.semanticscholar.org/CorpusID:57626488>
- Sotetsu Koyamada, Shinri Okano, Soichiro Nishimori, Yu Murata, Keigo Habara, Haruka Kita, and Shin Ishii. Pgx: Hardware-accelerated parallel game simulators for reinforcement learning. In *Advances in Neural Information Processing Systems*, volume 36, pp. 45716–45743, 2023.
- Mathias Lechner, Tim Seyde, Tsun-Hsuan Johnson Wang, Wei Xiao, Ramin Hasani, Joshua Rountree, Daniela Rus, et al. Gigastep-one billion steps per second multi-agent reinforcement learning. *Advances in Neural Information Processing Systems*, 36:155–170, 2023.
- Joel Lehman and Kenneth O Stanley. Abandoning objectives: Evolution through the search for novelty alone. *Evolutionary computation*, 19(2):189–223, 2011.
- Joel Z Leibo, Julien Perolat, Edward Hughes, Steven Wheelwright, Adam H Marblestone, Edgar Duñez-Guzmán, Peter Sunehag, Iain Dunning, and Thore Graepel. Malthusian reinforcement learning. *arXiv preprint arXiv:1812.07019*, 2018.
- Joel Z Leibo, Edward Hughes, Marc Lanctot, and Thore Graepel. Autocurricula and the emergence of innovation from social interaction: A manifesto for multi-agent intelligence research. *arXiv preprint arXiv:1903.00742*, 2019.

- Chris Lu, Michael Beukman, Michael Matthews, and Jakob Foerster. Jaxlife: An open-ended agentic simulator. In *Artificial Life Conference Proceedings 36*, volume 2024, pp. 47. MIT Press One Rogers Street, Cambridge, MA 02142-1209, USA journals-info . . . , 2024.
- Sean Luke, Claudio Cioffi-Revilla, Liviu Panait, Keith Sullivan, and Gabriel Balan. Mason: A multiagent simulation environment. *Simulation*, 81(7):517–527, 2005.
- Robert H MacArthur and Edward O Wilson. *The theory of island biogeography*, volume 1. Princeton university press, 2001.
- Horia Mania, Aurelia Guy, and Benjamin Recht. Simple random search provides a competitive approach to reinforcement learning. *arXiv preprint arXiv:1803.07055*, 2018.
- Michael Matthews, Michael Beukman, Benjamin Ellis, Mikayel Samvelyan, Matthew Jackson, Samuel Coward, and Jakob Foerster. Craftax: A lightning-fast benchmark for open-ended reinforcement learning. *arXiv preprint arXiv:2402.16801*, 2024.
- Melanie Mitchell, Michael D. Thomure, and Nathan L. Williams. The role of space in the success of coevolutionary learning. In *Proceedings of ALIFE X: The Tenth International Conference on the Simulation and Synthesis of Living Systems*, Bloomington, Indiana, USA, 2006.
- Igor Mordatch and Pieter Abbeel. Emergence of grounded compositional language in multi-agent populations. In *Proceedings of the AAAI conference on artificial intelligence*, volume 32, 2018.
- Alexander Mordvintsev, Ettore Randazzo, Eyvind Niklasson, and Michael Levin. Growing neural cellular automata. *Distill*, 2020. doi: 10.23915/distill.00023. <https://distill.pub/2020/growing-ca>.
- Ugurcan Mugan and Malcolm A MacIver. Spatial planning with long visual range benefits escape from visual predators in complex naturalistic environments. *Nature communications*, 11(1):3057, 2020.
- Joon Sung Park, Joseph O’Brien, Carrie Jun Cai, Meredith Ringel Morris, Percy Liang, and Michael S Bernstein. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th annual acm symposium on user interface software and technology*, pp. 1–22, 2023.
- Aleksei Petrenko, Erik Wijmans, Brennan Shacklett, and Vladlen Koltun. Megaverse: Simulating embodied agents at one million experiences per second. In *International Conference on Machine Learning*, pp. 8556–8566. PMLR, 2021.
- Sarah Pratt, Luca Weihs, and Ali Farhadi. The introspective agent: Interdependence of strategy, physiology, and sensing for embodied agents. *arXiv preprint arXiv:2201.00411*, 2022.
- Ettore Randazzo and Alexander Mordvintsev. Biomaker ca: a biome maker project using cellular automata. *arXiv preprint arXiv:2307.09320*, 2023.
- Thomas S Ray. Evolution, ecology and optimization of digital organisms. *Santa Fe*, 1992.
- Ingo Rechenberg. Evolutionsstrategie : Optimierung technischer systeme nach prinzipien der biologischen evolution. 1973. URL <https://api.semanticscholar.org/CorpusID:60975248>.
- Craig W Reynolds. Flocks, herds and schools: A distributed behavioral model. In *Proceedings of the 14th annual conference on Computer graphics and interactive techniques*, pp. 25–34, 1987.
- Paul Richmond, Robert Chisholm, Peter Heywood, Matthew Leach, and Mozhgan Kabiri Chimeh. Flame gpu, 2024. URL <https://zenodo.org/doi/10.5281/zenodo.5428984>.
- Max Robinson. Life engine. <https://github.com/MaxRobinsonTheGreat/LifeEngine>, 2021.
- Tim Salimans, Jonathan Ho, Xi Chen, and Ilya Sutskever. Evolution strategies as a scalable alternative to reinforcement learning. *ArXiv*, abs/1703.03864, 2017. URL <https://api.semanticscholar.org/CorpusID:11410889>.

- Dolph Schluter. *The Ecology of Adaptive Radiation*. Oxford University Press, Oxford, 2000.
- Hans-Paul Schwefel. Evolution and optimum seeking. In *Sixth-generation computer technology series*, 1995. URL <https://api.semanticscholar.org/CorpusID:43418053>
- George Gaylord Simpson. *Tempo and Mode in Evolution*. Columbia University Press, New York, 1944.
- Karl Sims. Evolving virtual creatures. *Proceedings of the 21st annual conference on Computer graphics and interactive techniques*, 1994a. URL <https://api.semanticscholar.org/CorpusID:28435>
- Karl Sims. Evolving 3d morphology and behavior by competition. *Artificial life*, 1(4):353–372, 1994b.
- Kenneth O Stanley and Risto Miikkulainen. Evolving neural networks through augmenting topologies. *Evolutionary computation*, 10(2):99–127, 2002.
- Kenneth O Stanley and Risto Miikkulainen. Competitive coevolution through evolutionary complexification. *Journal of artificial intelligence research*, 21:63–100, 2004.
- Kenneth O Stanley, David B D’Ambrosio, and Jason Gauci. A hypercube-based encoding for evolving large-scale neural networks. *Artificial life*, 15(2):185–212, 2009.
- Joseph Suarez, Yilun Du, Phillip Isola, and Igor Mordatch. Neural mmo: A massively multi-agent game environment for training and evaluating intelligent agents. *arXiv preprint arXiv:1903.00784*, 2019.
- Joseph Suarez, David Bloomin, Kyoung Whan Choe, Hao Xiang Li, Ryan Sullivan, Nishaanth Kanna, Daniel Scott, Rose Shuman, Herbie Bradley, Louis Castricato, et al. Neural mmo 2.0: a massively multi-task addition to massively multi-agent learning. *Advances in Neural Information Processing Systems*, 36:50094–50104, 2023.
- Felipe Petroski Such, Vashisht Madhavan, Edoardo Conti, Joel Lehman, Kenneth O Stanley, and Jeff Clune. Deep neuroevolution: Genetic algorithms are a competitive alternative for training deep neural networks for reinforcement learning. *arXiv preprint arXiv:1712.06567*, 2017.
- Richard S Sutton, Andrew G Barto, et al. *Reinforcement learning: An introduction*, volume 1. MIT press Cambridge, 1998.
- Paul Szerlip and Kenneth Stanley. Indirectly encoded sodarace for artificial life. In *Artificial Life Conference Proceedings*, pp. 218–225. Citeseer, 2013.
- Open Ended Learning Team, Adam Stooke, Anuj Mahajan, Catarina Barros, Charlie Deck, Jakob Bauer, Jakub Sygnowski, Maja Trebacz, Max Jaderberg, Michael Mathieu, et al. Open-ended learning leads to generally capable agents. *arXiv preprint arXiv:2107.12808*, 2021.
- Seth Tisue, Uri Wilensky, et al. Netlogo: A simple environment for modeling complexity. In *International conference on complex systems*, volume 21, pp. 16–21. Boston, MA, 2004.
- Kushagra Tiwary, Aaron Young, Zaid Tasneem, Tzofi Klinghoffer, Akshat Dave, Tomaso Poggio, Dan-Eric Nilsson, Brian Cheung, and Ramesh Raskar. What if eye...? computationally recreating vision evolution. *arXiv preprint arXiv:2501.15001*, 2025.
- Rui Wang, Joel Lehman, Jeff Clune, and Kenneth O Stanley. Paired open-ended trailblazer (poet): Endlessly generating increasingly complex and diverse learning environments and their solutions. *arXiv preprint arXiv:1901.01753*, 2019.
- Jason Wei, Yi Tay, Rishi Bommasani, Colin Raffel, Barret Zoph, Sebastian Borgeaud, Dani Yogatama, Maarten Bosma, Denny Zhou, Donald Metzler, et al. Emergent abilities of large language models. *arXiv preprint arXiv:2206.07682*, 2022a.

756 Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny
757 Zhou, et al. Chain-of-thought prompting elicits reasoning in large language models. *Advances in*
758 *neural information processing systems*, 35:24824–24837, 2022b.

759

760 Daan Wierstra, Tom Schaul, Tobias Glasmachers, Yi Sun, Jan Peters, and Jürgen Schmidhuber.
761 Natural evolution strategies. *The Journal of Machine Learning Research*, 15(1):949–980, 2014.

762

763 Sewall Wright. Evolution in mendelian populations. *Genetics*, 16(2):97–159, 1931.

764

765 Larry Yaeger et al. Computational genetics, physiology, metabolism, neural systems, learning, vi-
766 sion, and behavior or poly world: Life in a new context. In *SANTA FE INSTITUTE STUDIES*
767 *IN THE SCIENCES OF COMPLEXITY-PROCEEDINGS VOLUME-*, volume 17, pp. 263–263.
768 ADDISON-WESLEY PUBLISHING CO, 1994.

769

770 Jun Yamada, John Shawe-Taylor, and Zafeirios Fountas. Evolution of a complex predator-prey
771 ecosystem on large-scale multi-agent deep reinforcement learning. In *2020 International joint*
772 *conference on neural networks (IJCNN)*, pp. 1–8. IEEE, 2020.

773

774 Lianmin Zheng, Jiacheng Yang, Han Cai, Weinan Zhang, Jun Wang, and Yong Yu. Magent: A
775 many-agent reinforcement learning platform for artificial collective intelligence. *arXiv preprint*
776 *arXiv:1712.00600*, 2017.

777

778

779

780

781

782

783

784

785

786

787

788

789

790

791

792

793

794

795

796

797

798

799

800

801

802

803

804

805

806

807

808

809

A ADDITIONAL DATA

Table 3: Per-seed mining events in the **Beach** world for populations of agents with a compass (**RC**) versus without a compass (**R**), across five scales.  indicates whether a run exhibited a mining event.  reports the percentage decrease in free biomass on dry land during the mining event, and  the approximate timestep at which free dry biomass reached the local maximum marking the onset of mining. For runs with multiple mining events, we report the first.  gives the timestep at which the run went extinct, or n/a if it survived the full 2M steps.

	1024x1024 32768				512x512 8192				256x256 2048				128x128 512				64x64 128			
Compass On (RC)																				
Seed																				
0	✓	35k	11.1	n/a	✓	105k	33.	n/a	✓	145k	49.3	n/a	✓	105k	72.5	n/a	x			
1	✓	90k	16.2	n/a	✓	50k	17.2	n/a	✓	50k	46.6	n/a	x		140	✓	70k			
2	✓	40k	15.	n/a	✓	30k	14.3	n/a	✓	55k	34.7	n/a	✓	50k	50.7	n/a	✓			
3	✓	60k	15.	n/a	✓	45k	15.6	n/a	✓	90k	47.4	n/a	✓	20k	52.	n/a	x			
Avg	4/4	34k	14.3	0/4	4/4	58k	20.	0/4	4/4	85k	44.5	0/4	3/4	58k	58.4	1/4	2/4			
Compass Off (R)																				
Seed																				
0	x			n/a	x			945k	x			185k	x			65k	x			
1	✓	275k	12.4	n/a	x			1085k	x			110k	x			55k	x			
2	x			n/a	x			1070k	x			165k	x			55k	x			
3	x			n/a	x			280k	x			125k	x			90k	x			
Avg	1/4	275k	12.4	0/4	0/4			n/a	0/4			4/4	0/4			4/4	0/4			

Table 4: Per-seed mining events in the **Island**, **Lake**, **Isthmus**, and **Channel** terrains for **(RC)** agents versus **(R)** agents, at grid size 512×512 with an initial population of 8192.

	Island				Lake				Isthmus				Channel			
Compass On (RC)																
Seed																
0	✓	40k	94.3	n/a	x			45k	✓	45k	33.	n/a	✓	50k	27.4	n/a
1	✓	30k	94.3	n/a	✓	30k	22.5	n/a	✓	55k	37.3	n/a	✓	80k	30.8	n/a
2	✓	30k	94.8	n/a	x			45k	✓	40k	28.1	n/a	✓	50k	41.	n/a
3	✓	70k	95.1	n/a	✓	35k	25.3	n/a	✓	30k	41.5	n/a	x			80k
Avg	4/4	43k	94.6	0/4	2/4	33k	23.9	2/4	4/4	43k	35.	0/4	3/4	60k	33.1	1/4
Compass Off (R)																
Seed																
0	x			185k	x			125k	x			145k	x			100k
1	✓	75k	30.9	n/a	x			130k	x			260k	x			105k
2	✓	85k	47.	n/a	✓	75k	23.4	1400k	✓	95k	18.8	n/a	x			105k
3	✓	115k	39.5	n/a	✓	70k	24.3	1250k	x			140k	x			100k
Avg	3/4	92k	39.1	1/4	2/4	73k	23.9	4/4	1/4	95k	18.8	3/4	0/4			4/4

Table 5: Population and foraging data for Vision and Blind agents in the **Ocean** world at different scales. denotes population size; denotes biomass utilization, the percentage of the system's total biomass currently held in the agents' stomachs; denotes move actions per agent, the fraction of time an agent took a move action, averaged over agents; and denotes the corresponding proportion of eat actions. Reported values of , , , and are averaged over all steps of the seed. Across scales, vision agents sustain larger populations and higher biomass utilization while moving less and eating more, the quantitative counterpart of their foraging advantage.

	1024x1024 32768				512x512 8192				256x256 2048				128x128 512			
Vision (RCV+A)																
Seed																
0	58354	.396	.402	.426	16413	.369	.257	.532	4195	.382	.255	.536	1126	.426	.261	.541
1	65107	.365	.257	.529	16200	.364	.256	.533	4142	.377	.258	.536	1103	.415	.262	.537
2	64735	.363	.257	.530	16560	.372	.257	.533	4269	.388	.260	.536	1041	.392	.254	.542
3	64595	.362	.256	.532	16206	.365	.255	.531	4249	.388	.259	.533	1100	.414	.257	.538
Avg	63198	.371	.293	.504	16345	.368	.256	.532	4213	.384	.258	.535	1092	.412	.259	.539
Blind (RC+A)																
Seed																
0	22734	.143	.446	.372	6549	.173	.442	.400	2036	.209	.433	.461	545	.219	.465	.427
1	23670	.151	.426	.388	6939	.180	.438	.396	3107	.321	.390	.416	1082	.444	.416	.460
2	24745	.159	.358	.426	5448	.136	.474	.374	1587	.163	.440	.433	979	.421	.404	.436
3	20004	.130	.469	.359	10354	.265	.311	.439	3164	.324	.321	.450	788	.317	.431	.477
Avg	22788	.146	.425	.386	7323	.189	.416	.402	2473	.254	.396	.440	848	.351	.429	.450

Table 6: Attacking data for Vision and Blind agents in the **Ocean** world at different scales.  denotes attacks per agent, the fraction of time an agent took the attack action, averaged over all agents, and  denotes homicides per attack, the number of agents killed per attack action. Reported values of  and  are averaged over all steps of the seed. Vision agents combine a low attack rate with a high kill rate—the signature of targeted attacking—across every scale. The outlier of 0.562 for  in seed 3 of the 128×128 (**RC+A**) run reflects extinction before 2M steps, which spikes the attacks-per-agent metric as the population dwindles.

	1024x1024		512x512		256x256		128x128	
	32768		8192		2048		512	
Vision (RCV+A)								
Seed								
0	.011	.776	.015	.731	.013	.784	.010	.788
1	.016	.733	.015	.741	.014	.780	.011	.793
2	.016	.707	.015	.737	.014	.750	.012	.771
3	.016	.712	.015	.780	.013	.759	.011	.765
Avg	.015	.732	.015	.747	.014	.758	.011	.779
Blind (RC+A)								
Seed								
0	.070	.218	.042	.437	.005	1.074	.016	.558
1	.054	.300	.044	.383	.016	.871	.006	1.032
2	.076	.236	.048	.238	.031	.383	.010	.979
3	.059	.203	.044	.456	.033	.446	.562	.841
Avg	.065	.239	.045	.378	.021	.693	.148	.853

B DEMOGRAPHY

We quantify how the rates of extinction and of mining emergence depend on world size and on an agent’s access to global direction through the compass sensor. Extinction is defined as the population reaching zero individuals, and mining is defined exactly as in the main experiments (a 10% drop in free biomass on land).

Runs are stratified jointly by world size and compass access. We simulated 64 runs each of 128×128 without compass and 128×128 with compass, and 32 runs each of 512×512 without compass and 512×512 with compass. For each of the four resulting groups we fit a non-parametric Kaplan–Meier estimator to the time-to-event distribution, treating runs that did not experience the event within 2M steps as right-censored. As Figures 7 and 8 make clear, worlds survive longer and produce mining more rapidly when they are large or when agents are granted compass information—the two factors identified as decisive in the main text.

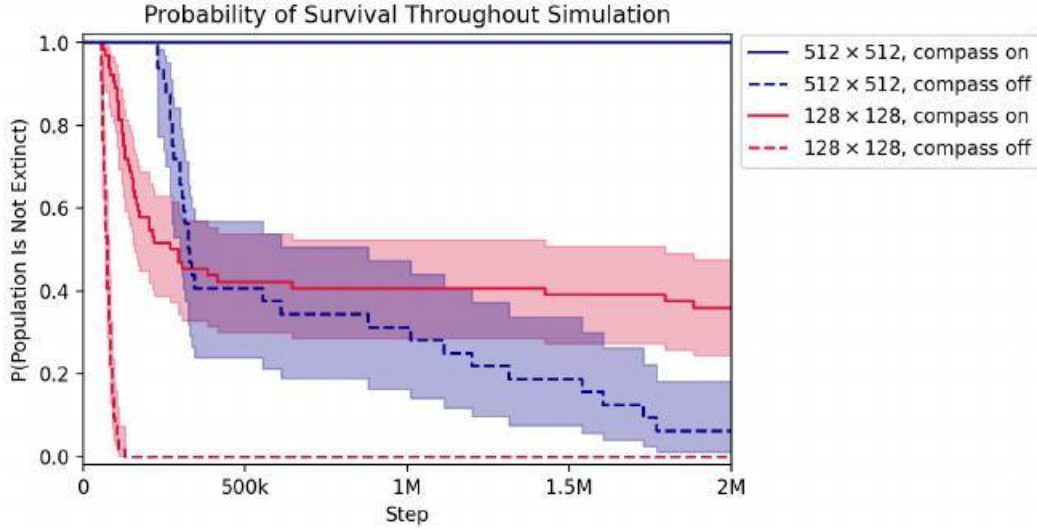


Figure 7: Probability that the population has not gone extinct, as a function of world size, compass access, and timestep. A separate non-parametric Kaplan–Meier estimator is fit to each group of runs (e.g., 512×512 , compass on), with shaded areas denoting the 95% confidence interval for each curve. Each 512×512 group is fit on 32 runs and each 128×128 group on 64 runs. Larger worlds and compass access both prolong survival.

We further compare emergence and extinction timing with a Cox proportional hazards model, a semi-parametric survival model relating covariates to event hazard rates (Table 7). Runs that experienced neither emergence nor extinction within 2M time steps were treated as censored. For extinction, both larger world size and compass use substantially reduce the hazard: the extinction rate at 128×128 is roughly 25 times that at 512×512 ($HR=0.04$), and enabling the compass yields an effect of comparable magnitude ($HR=0.03$). For mining emergence, larger worlds generate mining much more rapidly ($HR=6.07$), while compass use accelerates emergence dramatically ($HR=34.59$). Both models discriminate strongly between groups (concordance 0.80 for extinction, 0.85 for mining). These estimates are consistent with a single coherent mechanism: larger worlds sustain larger populations, which lowers the probability of reaching the absorbing zero-population boundary while simultaneously exploring a larger phenotype space through a greater aggregate quantity of mutation than smaller worlds afford.

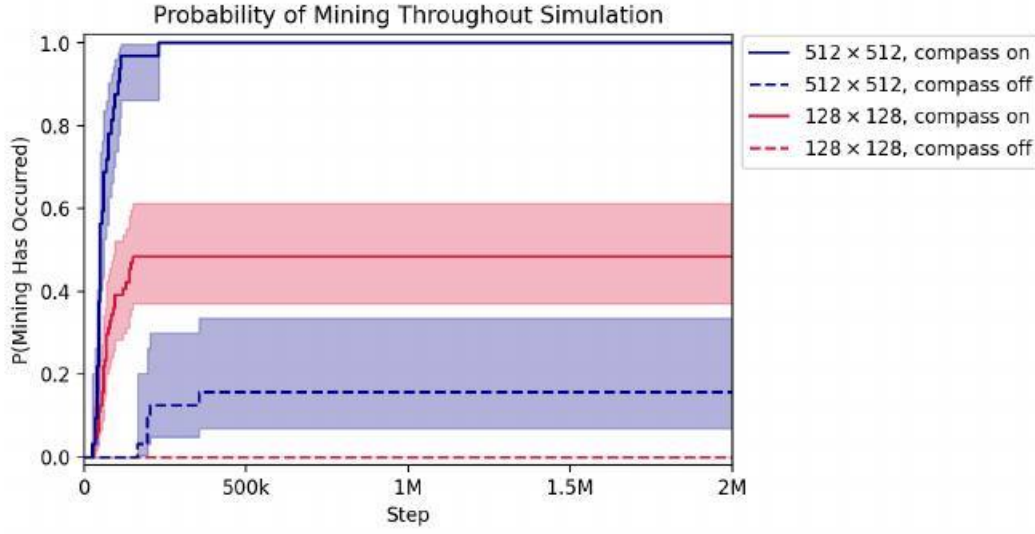


Figure 8: Probability that mining has evolved within the population, as a function of world size, compass access, and timestep. A separate non-parametric Kaplan–Meier estimator is fit to each group of runs (e.g., 512×512 , compass on), with shaded areas denoting the 95% confidence interval for each curve. Each 512×512 group is fit on 32 runs and each 128×128 group on 64 runs. Larger worlds and compass access both hasten and increase the emergence of mining.

Table 7: Cox proportional hazards models for extinction and emergence events, using a 128×128 , compass-off baseline. Hazard ratios (HR) below 1 indicate a reduced hazard (i.e., a longer time to event); for example, extinction $HR = 0.04$ for 512×512 indicates that extinction occurs at 0.04 times the rate observed at 128×128 . All four covariate effects are significant at $p < 0.005$. The analysis comprises 192 runs in total: 32 seeds with compass on and 32 with compass off in the 512×512 world, and 64 seeds with compass on and 64 with compass off in the 128×128 world.

Covariate	Extinction			Mining		
	HR	95% CI	p-value	HR	95% CI	p-value
Size 512×512	0.04	0.02–0.07	< 0.005	6.07	3.61–10.20	< 0.005
Compass On	0.03	0.01–0.05	< 0.005	34.59	13.51–88.57	< 0.005
Concordance	0.80			0.85		
Events / Total	135 / 192			68 / 192		

C ADDITIONAL FORAGING AND PREDATION RESULTS

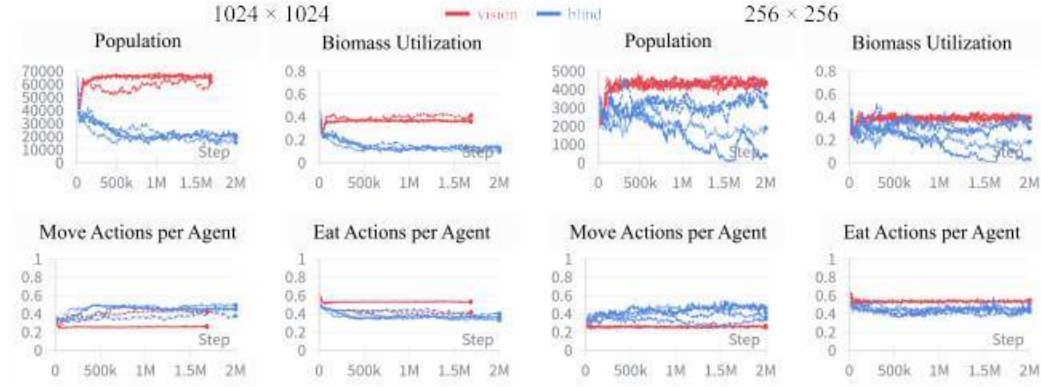


Figure 9: Foraging behavior in the **Ocean** world at different scales. Enabling vision (red) yields more stable populations and greater biomass utilization, driven by more frequent and more effective eating together with reduced movement. At larger scales these dynamics are more stable, with less variance across runs. Owing to limitations on GPU hours, the vision seeds at the 1024×1024 scale ran for 1.7M rather than 2M steps.

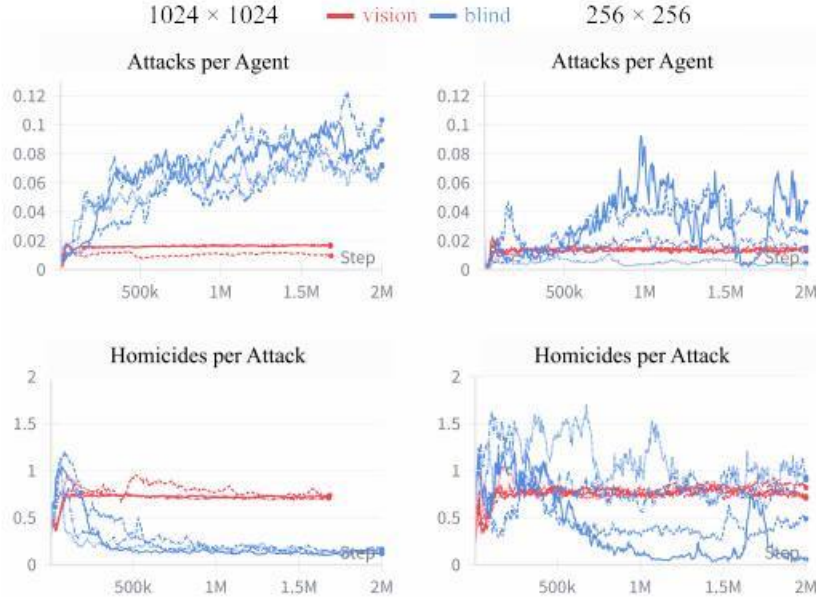


Figure 10: Attacking behavior in the **Ocean** world at different scales. With vision (red), agents attack less frequently but with substantially higher precision, consistent with the targeted, predation-like attacking that accompanies the foraging gains of Figure 9. This dynamic is again more pronounced at larger scales, with greater variance in the behavior of blind agents (blue) at smaller world sizes.

D ADDITIONAL EXPERIMENT VISUALIZATIONS

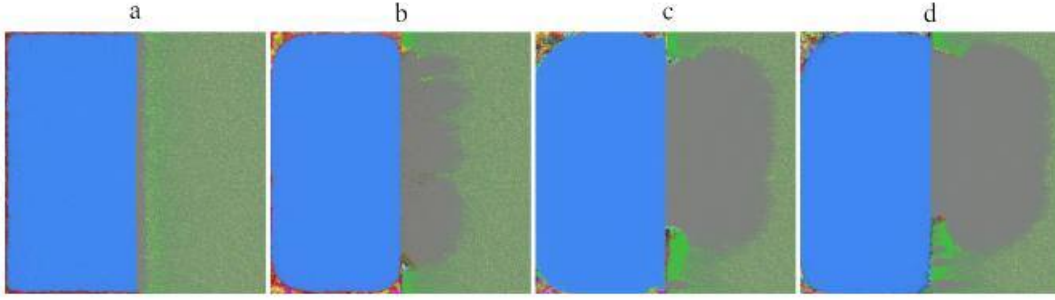


Figure 11: Larger-scale environments give rise to more varied ecologies and distinctive spatial patterns. In this 1024×1024 **Beach** world, (RC) agents initially died at the shore (a), transferring biomass onto dry land; they subsequently evolved mining behavior, clearing biomass from the beach (b). As biomass re-accumulated at the ends of the beach (c), subpopulations independently re-evolved mining, traveling inland along these secondary biomass strips to extract the resources (d)—an instance of the same strategy arising more than once within a single run.

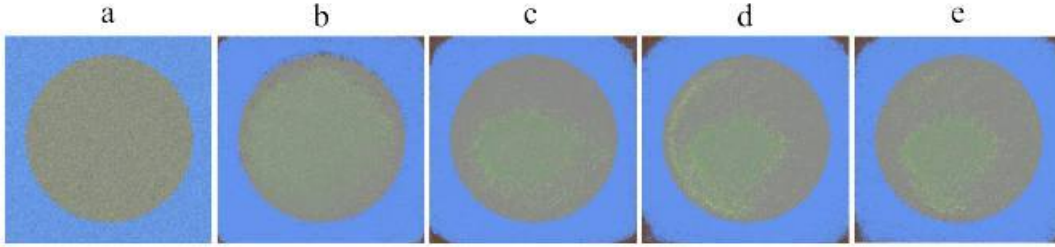


Figure 12: Mining behavior emerges across varied terrains. In this 512×512 **Island** world with (RC) agents, distinct regions of the shore are mined over the course of the run (b, c). Biomass is transferred back onto parts of the island once mining becomes difficult and agents die before returning to water (d), whereupon the population adapts to re-mine this accumulated wall of biomass, clearing it from the shore (e).

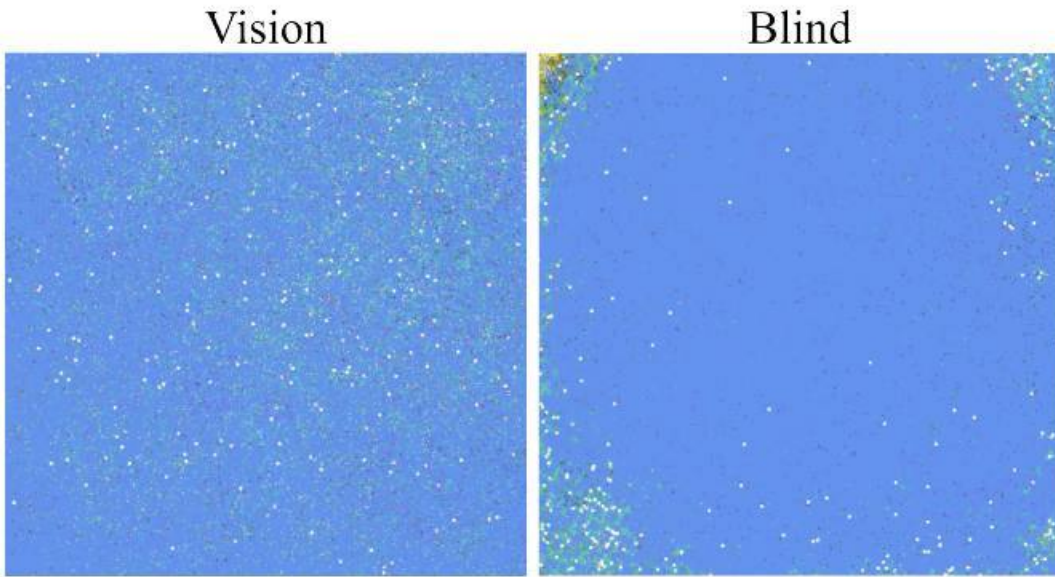


Figure 13: Representative runs in a 512×512 **Ocean** world with attacking enabled: vision (RCV+A) agents on the left and blind (RC+A) agents on the right. White squares mark locations where an agent is attacking. With vision, agents adapt to locate free resources throughout the ocean and to attack other agents far more effectively (see Figures 5 and 6). Blind agents explore for resources to a much lesser degree, instead adopting policies that cluster in the corners of the world.

E DETAILS ON COMPUTATION AND REPRODUCIBILITY

We detail the simulator design, runtime, and reproducibility considerations underlying our experiments.

Simulator design. All experiments are conducted with a JAX-based, GPU-accelerated simulator engineered specifically for large-scale multi-agent ecosystems with open-ended populations. The simulator computes every component of the environment step on the GPU—terrain dynamics, resource flows, agent updates, and measurement—so that policy evaluation and world simulation share a single JAX compilation pipeline. It exposes a deliberately minimal population-level interface: given the current map state, agent states, and heritable traits, a population algorithm returns actions and updated traits, while the simulator itself handles births, deaths, and environmental transitions. This separation of concerns allows a single environment implementation to be reused, without modification, across evolutionary algorithms, reinforcement learning methods, and other population update rules. Because all operations are implemented as pure JAX functions and combined through vectorization and just-in-time compilation, they fuse into a small number of GPU kernels, delivering the throughput that our scaling experiments demand; runtime measurements on a single H100 GPU are reported in Table 8.

A distinguishing feature of the simulator is its built-in tooling. A measurement API computes and records aggregate statistics at every step—biomass utilization, movement and eating rates, attack and homicide rates, and an extensive set of environment-level parameters—which feed directly into our figures and tables. An interactive 3D viewer, in addition, renders terrain, water, resources, and agents from generated reports and supports inspection of individual agents (Figure 1). These tools proved essential for discovering and validating qualitative phenomena—such as long-range resource mining and the spatial patterns of attacking—that are not apparent from scalar metrics alone.

From a high-performance computing standpoint, our simulator occupies a distinct niche among existing systems. General-purpose agent-based frameworks such as MASON (Luke et al., 2005), FLAME-GPU (Richmond et al., 2024), and Griddly (Bamford et al., 2020) offer high configurability but demand substantial environment engineering and do not directly target the open-ended, evolving populations at the scales we study; GPU-native multi-agent RL environments, conversely, emphasize fixed agent counts and task-specific reward structures. Our simulator unifies GPU-native execution, explicit support for births and deaths with mutable traits, and configurable resource and terrain dynamics, thereby sustaining open-ended populations of over 10^5 agents while remaining interoperable with diverse population algorithms. This combination constitutes a platform purpose-built for scaling ecological experiments of the kind presented here, and a durable foundation for future work on emergent behavior in large populations. Our simulator, together with the exact configurations, seeds, and scripts used to produce our results, will be released upon acceptance.

Runtime data. Table 8 reports runtime across grid sizes and maximum agent populations, with each run conducted on a single H100 GPU for a maximum of 2M steps. The table gives runtimes for (RC) and (RCV + A) agents with 2 MLP layers and 64 channels per layer ($\sim 22k$ parameters total); runtime varies with agent network size, the active sensor set, and the level of logging.

Our largest runs, at the 1024×1024 grid size, require ~ 10 hours in the **Beach** terrain and ~ 14 hours in the **Ocean** terrain on a single H100, with memory usage of roughly 60GB. **Ocean** runs are somewhat slower owing to the enabled attack action and vision sensing. That populations of this magnitude can be simulated for millions of steps within these budgets is precisely what renders our scaling study feasible on a modest allocation of hardware.

Reproducibility. Because our environments are large and heavily randomized, the precise outcome of an experiment can vary substantially from run to run; we therefore report four distinct random seeds per experiment, and considerably more in our demographic analyses. Owing to nondeterminism in certain low-level GPU operations, repeating an experiment with the same random seed does not reliably reproduce an identical trajectory. Specifically, modern GPUs may execute parallel floating-point operations such as summations in different orders across runs, and may fuse operations in ways that alter rounding, so that minute discrepancies arising from finite-precision arithmetic accumulate over many steps into appreciable drift in the simulation trajectory. The use of 16-bit weights coarsens these rounding effects but is not their primary source. Notwithstanding this run-to-run variation, the high-level patterns we report emerge robustly once the results of multiple

Table 8: Runtime in the **Beach** and **Ocean** worlds across grid sizes and maximum agent populations. Each (**RC**) agent has a 22,465-parameter network and each (**RCV + A**) agent a 22,529-parameter network. We report simulation steps per second (hz) together with the total extrapolated wall-clock runtime in hours for 2M steps; the effective runtime is lower for seeds whose populations go extinct before 2M steps.

	1024 × 1024		512 × 512		256 × 256		128 × 128		64 × 64	
	131072		32768		16384		8192		4096	
Environment	hz	runtime	hz	runtime	hz	runtime	hz	runtime	hz	runtime
Beach (RC)	55	10.1 hrs	190	2.9 hrs	315	1.8 hrs	535	1.0 hrs	750	0.7 hrs
Ocean (RCV + A)	40	13.9 hrs	142	3.9 hrs	235	2.4 hrs	415	1.3 hrs	750	0.7 hrs

seeds are aggregated, which is precisely why our conclusions are stated at the level of aggregate frequencies and hazard rates rather than individual trajectories.

F ABLATIONS

We present a series of ablations that characterize the effect of individual hyperparameters and establish the robustness of our results across varying environment and agent configurations.

F.1 INITIAL POPULATION SIZE SWEEP FOR FIXED MAP SIZE

We first evaluate how initial population size affects the consistency of mining behavior and the stability of the population in a 256×256 **Beach** world, reporting results for (**RC**) agent populations. As a baseline, agents without a compass (**R**) exhibited a 0% mining rate and a 100% extinction rate across all initial population sizes except the largest tested (8192), which yielded 25% mining and 75% extinction. We note that mining and extinction frequencies need not sum to 100%: a population may, for instance, develop mining and subsequently go extinct, or persist for the full 2M steps without ever developing mining.

As Figure 14 shows, initial population size has no appreciable effect on either mining consistency or extinction rate: populations seeded with only 128 agents evolve mining as frequently as those seeded with 8192, and extinction rates remain largely invariant across the entire range.

Figure 15 clarifies why this effect is so muted. Irrespective of its initial size, each run converges to a comparable population within the first 50,000 steps, indicating that the environment implicitly imposes a carrying capacity fixed by the resources available in the system: populations beginning below capacity grow to meet it, while populations exceeding it are driven back down. Once the initial size has been corrected early in the run, the effective population is therefore similar across conditions, leaving mining and extinction frequencies unaffected. Larger initial populations do contribute modestly to the total resources in the system, since each agent carries biomass that is deposited on death; they accordingly attain slightly higher maximum populations on average over a run (Table 9), though the effect is too small to alter mining or extinction frequency in this setting. Finally, although population size converges early, it does not remain constant thereafter: later in a run, populations across different seeds can diverge substantially, as the ecological consequences of agents evolving distinct abilities at distinct times reshape the subsequent resource distribution.

F.2 VARYING RESOURCE RULES

We next evaluate the effect of two resource hyperparameters—the biomass photosynthesis rate and the initial biomass site density—on mining and extinction frequency. The setting is identical to that of Appendix F.1 with the initial population size fixed at 2048 and agents again equipped with a compass (**RC**).

The biomass photosynthesis rate governs the rate at which landscape biomass creates energy via photosynthesis, supplying resources to agents over time, while the initial biomass site density fixes

1296
1297
1298
1299
1300
1301
1302
1303
1304
1305
1306
1307
1308
1309
1310
1311
1312
1313
1314
1315
1316
1317
1318
1319
1320
1321
1322
1323
1324
1325
1326
1327
1328
1329
1330
1331
1332
1333
1334
1335
1336
1337
1338
1339
1340
1341
1342
1343
1344
1345
1346
1347
1348
1349

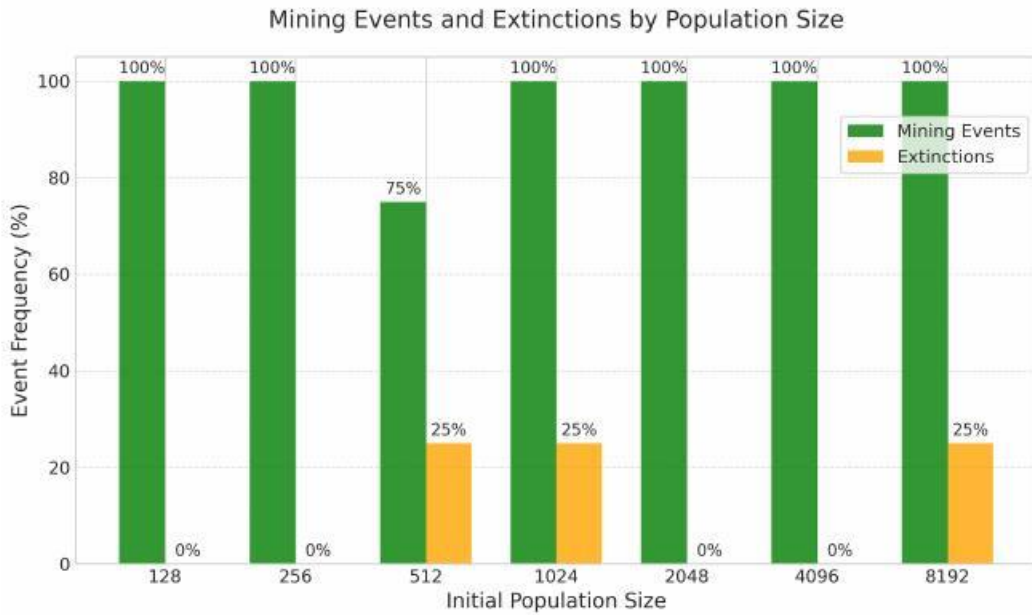


Figure 14: Frequency of mining and extinction events in a 256×256 **Beach** world across different initial sizes of (**RC**) agent populations. Frequencies are computed over four random seeds per initial population size; for example, a 25% extinction frequency means the population reached zero before 2M simulation steps in 1 of the 4 seeds. Mining consistency and extinction rate are essentially flat across the range of initial sizes.

Table 9: Maximum population size reached after step 0 across different starting population sizes, reported as the mean and standard deviation over four random seeds. Larger initial populations attain slightly higher maximum populations, and with greater cross-seed variance, than smaller ones. Nevertheless, the maximum population reached is largely robust to the initial size: an initial population of 128—only 1.6% of an initial size of 8192—reaches 78.9% of the maximum attained by the latter (3178 vs. 4026).

Initial Population Size	Max Population Size After Step 0
128	3178 ± 217
256	3056 ± 61
512	3451 ± 437
1024	3613 ± 381
2048	3786 ± 431
4096	3689 ± 437
8192	4026 ± 642

the fraction of grid cells containing biomass at the start of simulation. All main-body experiments (Section 4) use a biomass photosynthesis rate of 0.01 and an initial biomass site density of $1/4$.

Whereas varying the initial population size left mining and extinction consistency unchanged (Figure 14), varying the resource rules exerts a pronounced effect, as Figures 16 and 17 demonstrate.

The mechanism is direct: a higher biomass photosynthesis rate or initial biomass site density raises the effective carrying capacity of the environment by increasing the resources available in the system. Larger populations are thereby sustained throughout the run, which raises the likelihood of mutations favoring mining while lowering the probability of extinction. Our data bear this out: on average (omitting seeds that went extinct), runs with initial biomass site density $1/16$ reached a max-

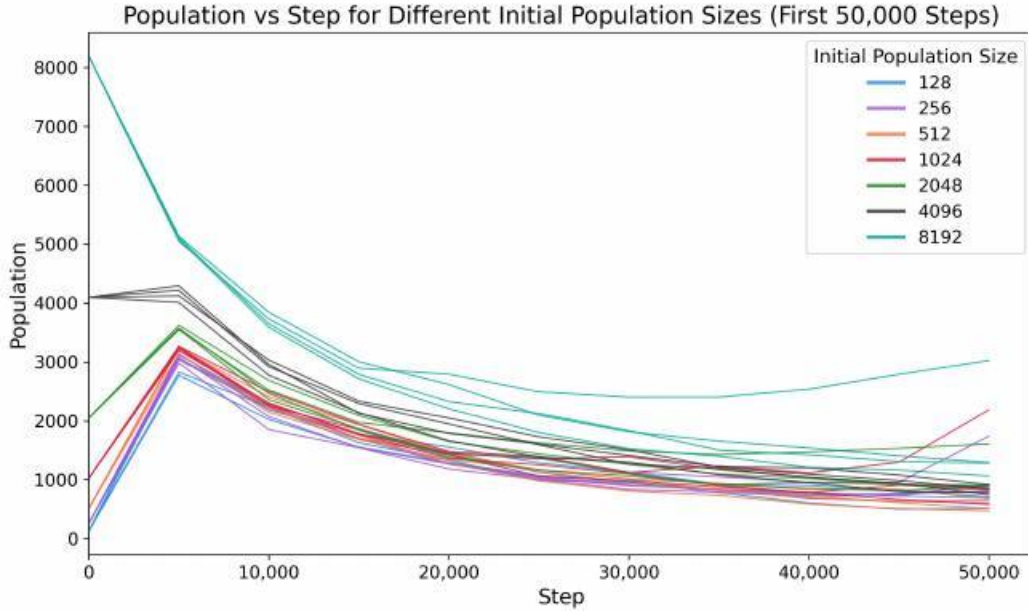


Figure 15: Population size during the first 50,000 time steps in the 256×256 **Beach** with (RC) agents, with four seeds per initial population size. The environment supports an implicit carrying capacity fixed by the resources available in the system, and every initial population size approaches this value early in the run. Larger initial populations contribute to the total resources in the system through biomass deposited on agent death, raising the carrying capacity somewhat.

imum post-step-0 population of 1608, rising to 2466 at density 1/8, 3786 at density 1/4, and 5353 at density 1/2.

F.3 RULING OUT SELF-AVERAGING IN LARGER POPULATIONS

One might object that the increased stability of mining in large environments (Table I) reflects nothing more than self-averaging, rather than qualitatively different dynamics. By self-averaging we mean the statistical-physics intuition that, as system size grows, fluctuations in macroscopic quantities shrink roughly as $1/\sqrt{N}$ —where N denotes the system size—so that a single large system behaves like the average of many smaller systems of the same total size. On this view, a 512×512 **Beach** world would amount to nothing more than a collection of loosely coupled 128×128 worlds run in parallel, and our scaling results could in principle be reproduced simply by running enough small worlds and averaging over random seeds. We now rule out this explanation directly.

We test the hypothesis by holding the total “sample size” approximately fixed while varying how it is partitioned across environments. In the **Beach** terrain with compass-equipped agents (RC), we consider three conditions:

- 4 runs of a 512×512 world with 8192 initial agents each,
- 16 runs of a 256×256 world with 2048 initial agents each,
- 64 runs of a 128×128 world with 512 initial agents each.

In each case, the total number of grid cells across runs, the total starting resources and the total initial number of initial agents are identical:

$$4 \times 512^2 = 16 \times 256^2 = 64 \times 128^2, \quad 4 \times 8192 = 16 \times 2048 = 64 \times 512.$$

Each seed is run for 2M steps with the same hyperparameters as in Section 4.1 and as a control we repeat the protocol for agents without a compass (R). Table 10 summarizes the frequency of long-distance mining and of extinction in each configuration.

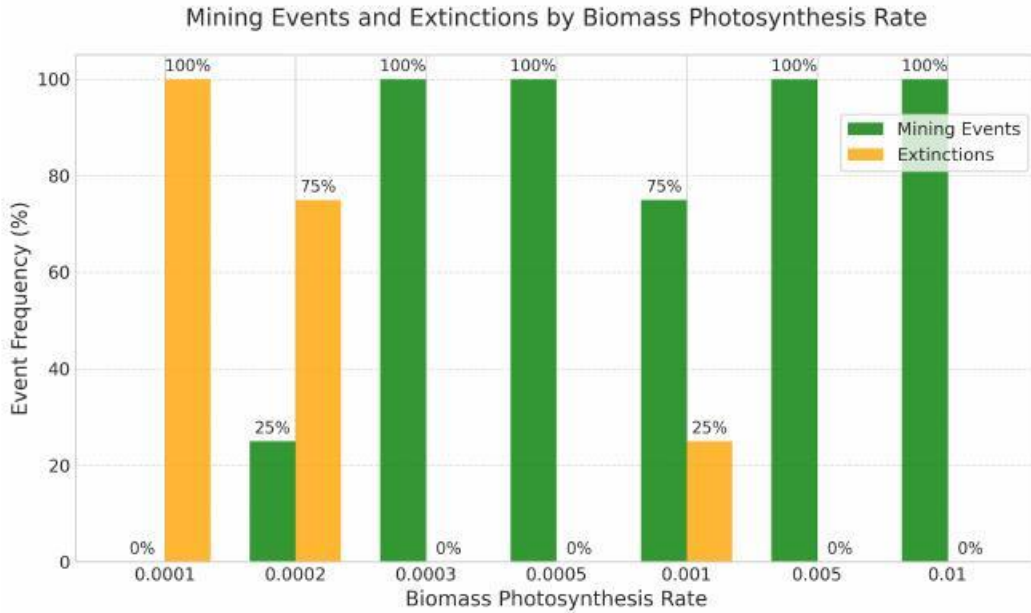


Figure 16: Frequency of mining and extinction events in a 256×256 **Beach** world across settings of the biomass photosynthesis rate. Frequencies are computed over four random seeds per setting. Mining emerges consistently once the biomass photosynthesis rate reaches 0.0003, consistent with the carrying-capacity mechanism described in the text.

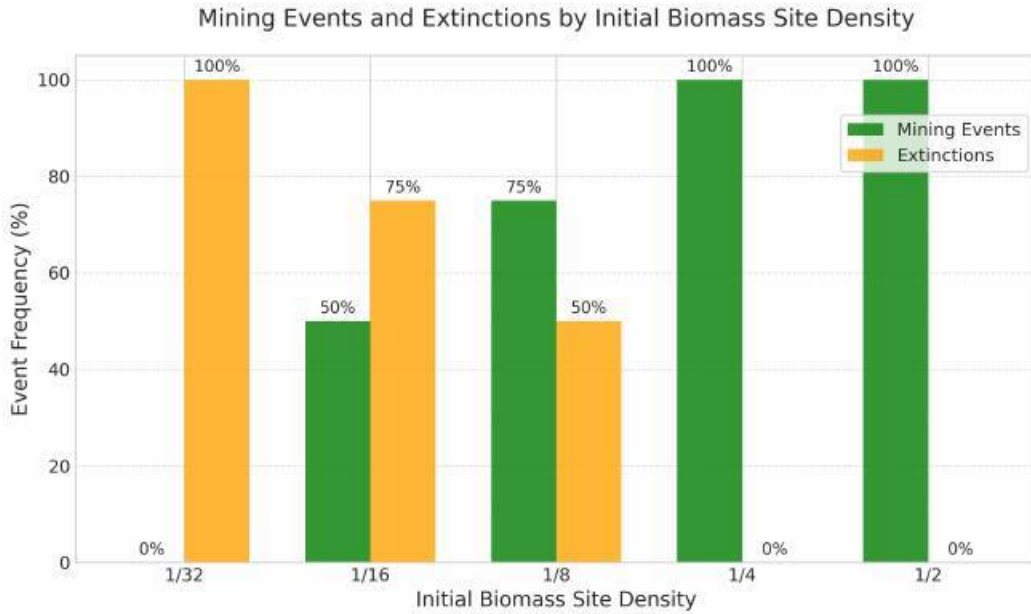


Figure 17: Frequency of mining and extinction events in a 256×256 **Beach** world across settings of the initial biomass site density. Frequencies are computed over four random seeds per setting. Higher initial biomass site densities yield more consistent discovery of mining together with a lower chance of extinction, again consistent with the carrying-capacity mechanism.

Under a self-averaging account, mining and extinction rates should be approximately invariant to how grid cells and agents are partitioned across runs, so that 4 large worlds and 64 small ones would exhibit similar probabilities of discovering mining. We observe instead a strong dependence on

world size. For **(RC)** agents, mining emerges in every run with no extinctions at 512×512 , remains common but no longer guaranteed at 256×256 , and becomes unreliable and fragile at 128×128 , where many populations go extinct before reaching 2M steps. **(R)** agents, by contrast, never develop mining and go extinct at every scale, confirming that the compass remains a key enabler of long-range behavior.

Table 10: Mining and extinction events when trading world size off against the number of independent runs. We compare **Beach** worlds at grid sizes 512×512 , 256×256 , and 128×128 with initial populations scaled proportionally (8192, 2048, and 512 agents, respectively). For each size we hold the total number of grid cells and initial agents fixed across all runs, using 4 seeds at 512×512 , 16 seeds at 256×256 , and 64 seeds at 128×128 . We report the fraction of runs in which mining emerged (🔍) and the fraction in which the population went extinct before 2M steps (💀), for agents with a compass **(RC)** and without a compass **(R)**. Mining consistency depends strongly on world size even though the aggregate cell and agent budget is held fixed, contradicting a self-averaging account.

	512×512 8192		256×256 2048		128×128 512	
	🔍	💀	🔍	💀	🔍	💀
(RC)	4/4	0/4	15/16	1/16	31/64	41/64
(R)	0/4	4/4	0/16	16/16	0/64	64/64

These results are incompatible with a self-averaging explanation; on the contrary, larger environments demonstrably enable and stabilize mining in ways that averaging over more small simulations cannot reproduce.

Even when the total number of cells, agents, and simulation steps is matched across conditions, concentrating resources into a few large worlds produces a regime in which mining is consistent and extinctions rare, whereas dispersing the same resources across many small worlds yields populations that frequently fail to discover mining and are far more prone to collapse. Increasing environmental scale therefore does not merely smooth out stochastic fluctuations; it alters the underlying ecological dynamics—for instance, by supporting larger, more spatially distributed populations and by sustaining more persistent diversity among evolving policies.

F.4 MUTATION RATE SWEEP

We assess the sensitivity of mining emergence to the mutation rate, the hyperparameter that sets the scale of the Gaussian noise added to a child’s policy at birth and thus governs how much a child differs from its parent. We again run experiments in a 256×256 **Beach** world with an initial population of 2048 **(RC)** agents for a maximum of 2M steps.

As Figure 18 shows, mining emerges in at least 3 of the 4 seeds for every mutation rate between 6×10^{-3} and 7.875×10^{-2} , establishing that the Section 4.1 results at mutation rate 3×10^{-2} are robust across more than an order of magnitude. Outside this interval the behavior degrades as expected: rates that are too small (e.g., 3×10^{-3}) prevent agents from developing new behavior and impair survival, culminating in extinction, whereas excessively large rates (e.g., 3×10^{-1}) destabilize the population, consistently producing extinction with no mining adaptation.

F.5 NETWORK SIZE

We study how the size and architecture of the agent policy network affect mining and extinction frequency, and how environmental scale modulates that effect. We examine 256×256 and 512×512 **Beach** worlds with initial populations of 2048 and 8192 **(RC)** agents, respectively (scaling by four with the number of grid cells), and run **(R)** baselines for each experiment—again observing rare to no mining events and 100% extinction rates.

The main experiments in Section 4 used agent policies of 2 layers and 64 channels, yielding 22465 parameters per agent (Table 11). As Figure 19 shows, in the 256×256 **Beach** halving the channels

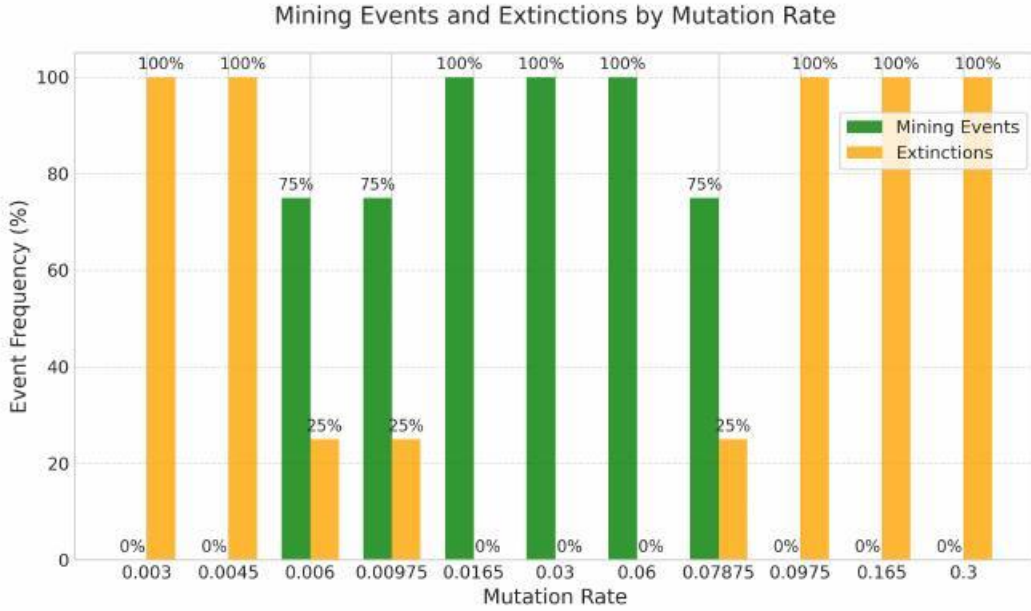


Figure 18: Frequency of mining and extinction events in a 256×256 **Beach** world across mutation rates. Frequencies are computed over four random seeds per setting. Mining is robust across the range 6×10^{-3} to 7.875×10^{-2} and degrades only at the extremes.

to 32 (9185 parameters) or doubling them to 128 (61313 parameters) at fixed depth has minimal to no effect on mining and extinction frequency. Network depth, by contrast, matters decisively: every population of 1-layer, 32-channel agents went extinct, as did 50% of populations of 3-layer, 128-channel agents.

Crucially, scaling the environment to the 512×512 grid size produces markedly different dynamics across network depths (Figure 20). In the larger world, populations with 3-layer, 128-channel networks consistently mined and survived through 2M steps and, strikingly, populations with the small 1-layer, 32-channel networks mined consistently as well. We hypothesize that both smaller and larger networks find it difficult to discover new behavior, though for distinct reasons: a smaller network’s reduced capacity makes good solutions hard to find, whereas a larger network’s expanded search space lengthens the time required to locate promising behavior. Whatever the precise mechanism, the observation that these networks fail so frequently in small environments yet succeed in larger ones further illustrates the stability conferred by scale. A larger environment supports larger populations, and thereby greater exploration of policy space; at the same time, locally successful mutants take longer to propagate across a large environment, which sustains diversity and again promotes exploration. For further discussion of these effects, see (Mitchell et al. 2006).

Table 11: Total parameter counts for (RC) and (R) agent policies across the network architectures we evaluate, specified by (layers, channels).

Architecture (layers, channels)	Parameter Count	
	(RC)	(R)
1, 32	8129	7969
2, 32	9185	9025
2, 64	22465	22145
2, 128	61313	60673
3, 128	77825	77185

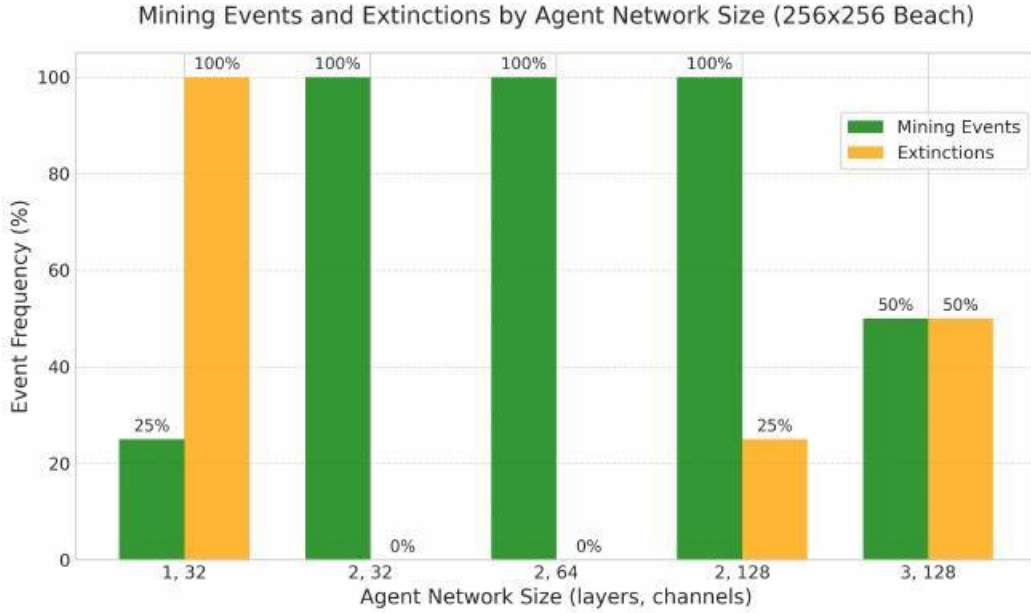


Figure 19: Frequency of mining and extinction events in a 256×256 **Beach** world across agent policy network sizes. Frequencies are computed over four random seeds per setting. At this scale, network depth is decisive whereas width is largely inconsequential.

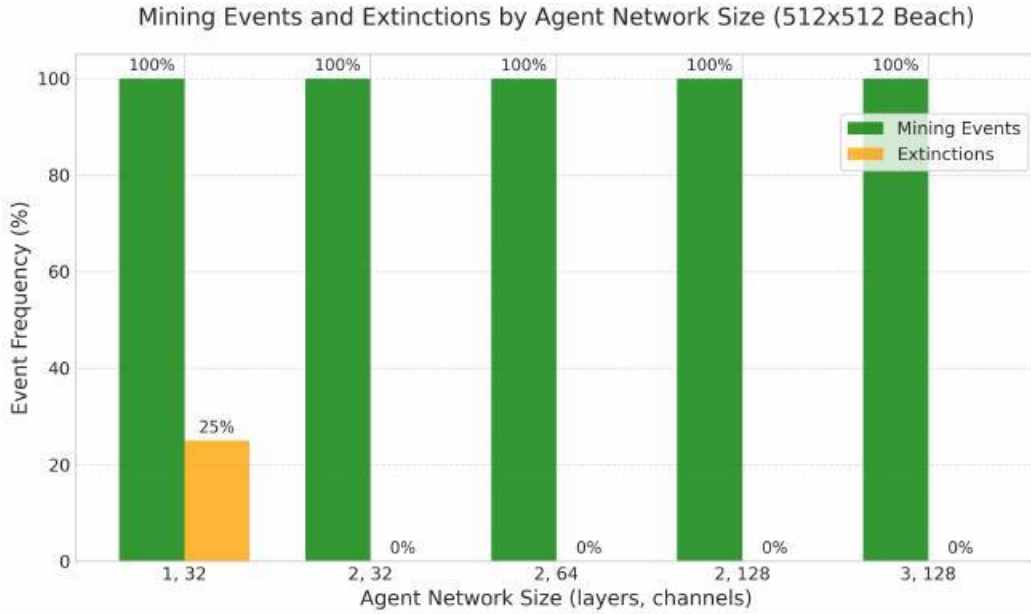


Figure 20: Frequency of mining and extinction events in a 512×512 **Beach** world across agent policy network sizes. Frequencies are computed over four random seeds per setting. In this larger world both shallow and deep networks mine consistently, in contrast to the 256×256 world of Figure

19

F.6 VARYING VISION RANGE

We assess how vision range and viewing configuration affect foraging and attacking behavior in the 512×512 **Ocean** world with (RCV+A) agents. We evaluate visual sensor ranges of 3×3 , 7×7 , and

9×9, and for each range consider the two viewing configurations shown in Figure 21 (a) Around, in which an agent sees grid cells in all directions, and (b) Front, in which an agent cannot see the cells behind it. The two configurations expose the agent to an equal total number of visible cells, isolating the effect of their spatial arrangement. The main vision experiments in Section 4.2 used the 7×7 Around setting.

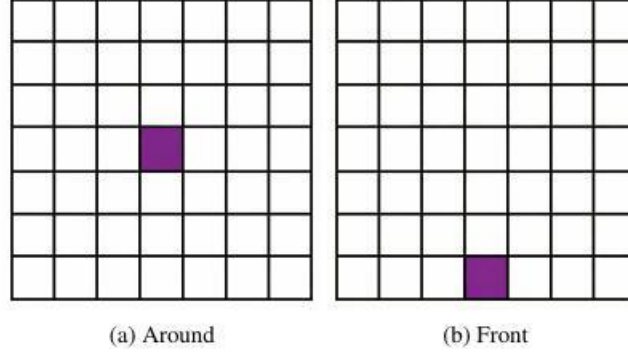


Figure 21: Comparison of the Around and Front viewing configurations at the 7×7 vision range. The purple cell denotes an agent facing the top of the page. Both configurations expose the agent to the same number of visible cells, differing only in their spatial arrangement.

Table 12: Foraging and attacking data in the 512×512 **Ocean** world for vision agents (**RCV+A**) across vision ranges and viewing configurations. See Tables 5 and 6 for definitions of the metrics denoted by each icon. For each seed, every metric is averaged over all steps from step 500k onward, allowing the metrics to stabilize; reported values are the mean across the four seeds together with the standard deviation of the four per-seed averages.

Vision Range	Viewing Configuration						
3×3	Blind	7212 ± 2496	.181 ± .060	.435 ± .090	.392 ± .033	.032 ± .006	.307 ± .090
	Around	10308 ± 4016	.258 ± .105	.352 ± .014	.427 ± .022	.040 ± .030	.388 ± .195
	Front	13863 ± 520	.351 ± .023	.336 ± .008	.437 ± .009	.026 ± .006	.470 ± .145
7×7	Around	16739 ± 122	.373 ± .003	.259 ± .000	.532 ± .001	.015 ± .000	.751 ± .007
	Front	15255 ± 623	.343 ± .015	.253 ± .001	.540 ± .004	.015 ± .000	.690 ± .041
9×9	Around	16744 ± 117	.372 ± .002	.257 ± .001	.534 ± .001	.015 ± .000	.728 ± .009
	Front	15211 ± 1416	.342 ± .033	.253 ± .001	.541 ± .007	.015 ± .000	.664 ± .050

These results indicate that the smaller 3 × 3 vision range yields behavior largely indistinguishable from having no vision at all (Blind), whereas the 9 × 9 range performs comparably to 7 × 7, pointing to diminishing returns from larger vision ranges in the present environmental configuration. This is intuitive: because the agents are memoryless, visual features at the periphery of a 9 × 9 window rarely alter the optimal action relative to a 7 × 7 window consistently enough to furnish a strong learning signal.

The data further suggest that, at the 7×7 and 9×9 ranges, the Around configuration is slightly more favorable for population stability and for the effectiveness of the attack and eat actions, producing higher average population , biomass utilization , and homicides per attack  than the Front configuration. We again attribute this to diminishing returns from more distant cells: under Around a greater share of the visible cells lie nearby, likely carrying more information relevant to near-term actions than the distant cells visible under Front. At the 3 × 3 range, however, the trend reverses and Front outperforms Around, indicating that here the marginal benefit of seeing one additional cell ahead exceeds that of seeing one cell behind—the latter yielding behavior that most closely resembles the Blind setting.

F.7 VARYING ATTACK STRENGTH

We study how reducing agent attack strength affects emergent behavior and population dynamics in the 256×256 **Ocean** world, evaluating three settings: (1) Full Attack, dealing 10 HP of damage and instantly killing any agent within the attack radius; (2) Half Attack, dealing 5 HP; and (3) Quarter Attack, dealing 2.5 HP. The main **Ocean** experiments in Section 4.2 used the Full Attack setting.

Reducing attack strength from Full to Half markedly diminishes both the effectiveness and the frequency of attacks, with attacks per agent and homicides per attack both dropping considerably (Figure 22). As a consequence, lower-attack-strength populations grow larger, with agents moving and eating more frequently and thereby achieving greater biomass utilization. Further reduction to the Quarter Attack setting yields diminishing returns, with little additional decrease in violence or growth in population. The mechanism is a compounding one: when each attack does less damage, a kill requires more attacks, so any single attack is less lethal; and as lethality falls, attacking becomes less useful and is selected for less strongly, so that agents at lower attack strengths never become as skillful at attacking and instead devote more time to foraging.

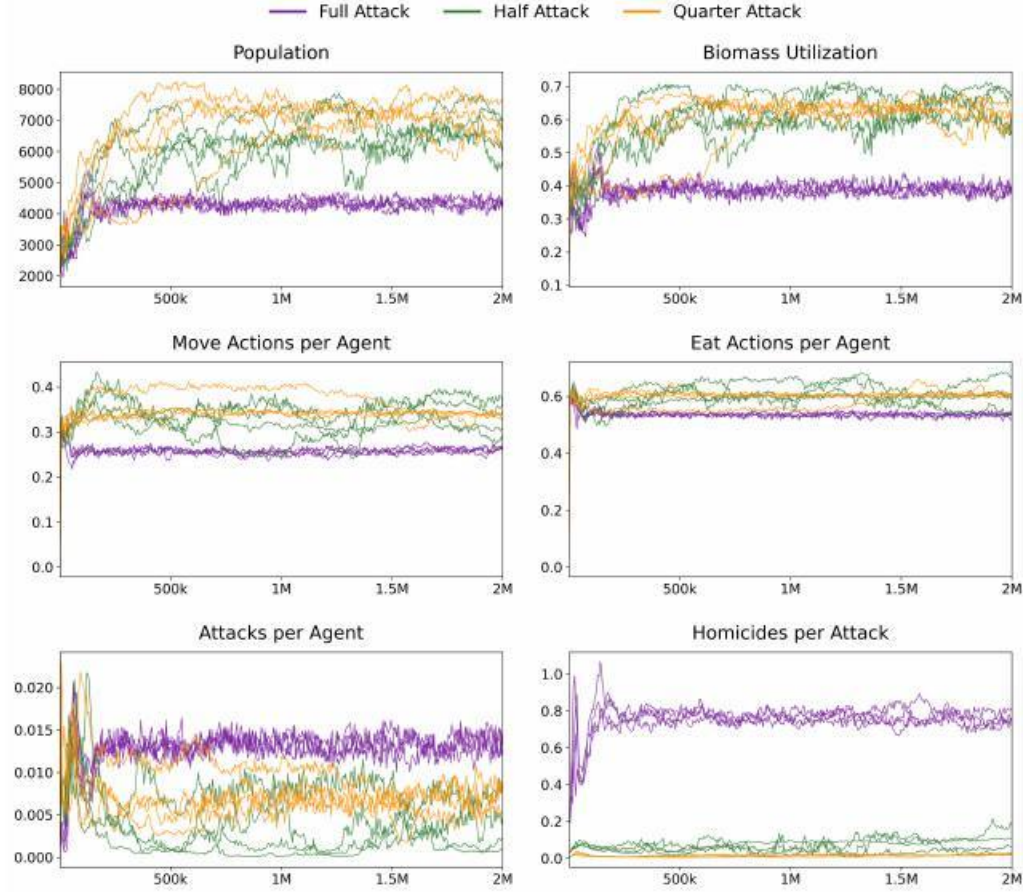


Figure 22: Population and action-frequency data over 2M steps in a 256×256 **Ocean** world for (RCV+A) agents across the Full, Half, and Quarter Attack strength settings, with four seeds plotted per setting. Weaker attacks yield less frequent and less lethal attacking together with larger, more actively foraging populations.

F.8 VARYING AGENT HP

We investigate the effect of reducing the starting HP of child agents and lowering the healing rate, thereby imposing pressure for agents to ‘grow’ over their lifetime in order to survive. The healing

rate determines how much HP an agent regenerates per healing step by converting available energy into HP, so that an agent can influence its own healing only by maintaining sufficient energy.

We study two settings: (1) Full HP, in which child HP starts at 5 with a healing rate of 1 HP per 0.1 energy, and (2) Low HP, in which child HP starts at 1 with a healing rate of 0.5. Maximum agent HP is 10 in all settings, and the main experiments (Section 4) used the Full HP setting.

Across the two settings we observe no appreciable difference in homicides per attack, eat-action frequency, or biomass utilization (Figure 23). Populations grow slightly larger under Full HP, whereas under Low HP agents attack less frequently and move slightly more. We hypothesize that agents born with lower HP must prioritize foraging early in life in order to grow to full size, which would account for the difference.

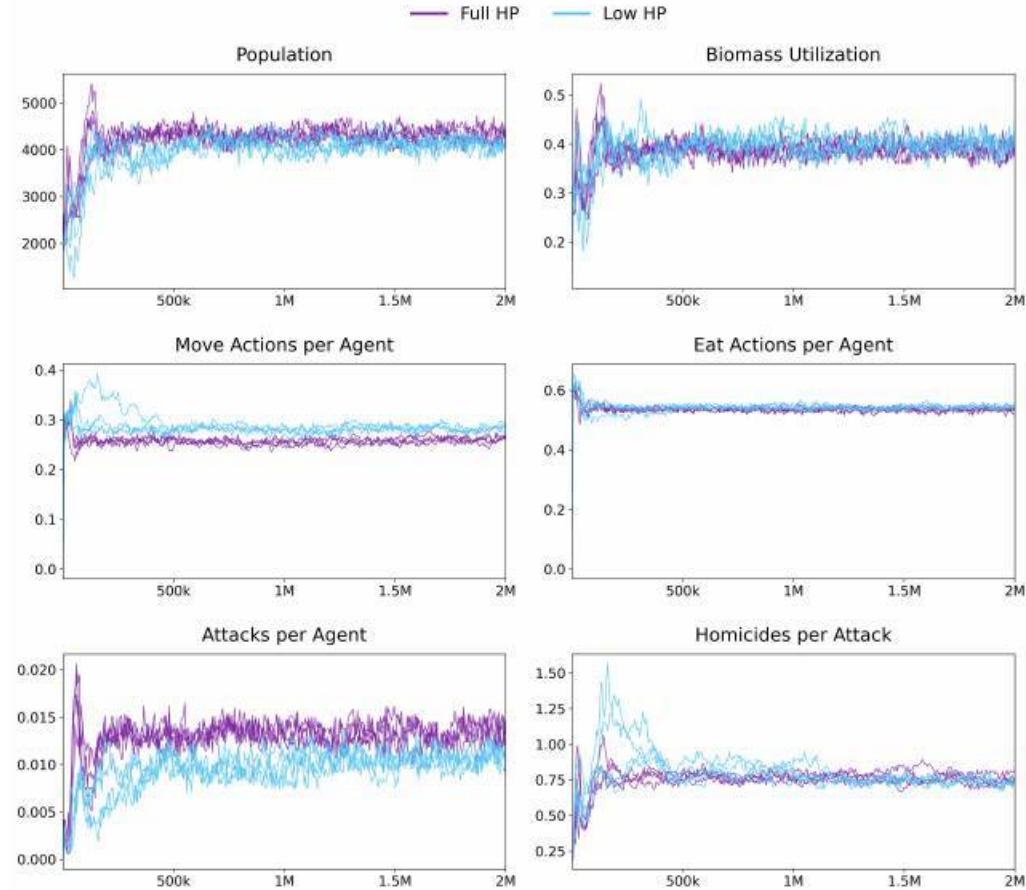


Figure 23: Population and action-frequency data over 2M steps in the 256×256 **Ocean** world for (RCV+A) agents across the Full HP and Low HP settings, with four seeds plotted per setting. Foraging and attacking effectiveness are largely unchanged; the principal differences are a slightly larger population under Full HP and slightly more movement under Low HP.